DESIGN REFERENCE GUIDE

EXISTING NON-RESIDENTIAL BUILDING

Version 4.0, February 2026

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1. About GreenRE

GreenRE Sdn Bhd is a wholly owned subsidiary of the Real Estate and Housing Developers' Association (REHDA). The GreenRE rating tool has been developed for the purposes as mentioned herein and may be subject to updating and/or modification in the future.

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GreenRE Sdn Bhd
Wisma REHDA,
No. 2C, Jalan SS5/6D,
47301 Petaling Jaya,
Selangor, Malaysia

2. Introduction

The GreenRE assessment scheme was established in 2013 and is a recognised green building rating system tailored for the tropical climate. GreenRE sets parameters and establishes indicators to guide the design, construction and operation of buildings towards increased energy effectiveness and enhanced environmental performance.

The intent of this Design Reference Guide for Existing Non-Residential Buildings (referred to as "this Guideline") is to establish environmentally friendly practices for the planning, design, construction and operation of buildings, which would help to mitigate the environmental impact of built structures.

This Guideline is not intended to abridge safety, health, environmental or related requirements contained in other applicable laws, codes or policies administered by relevant authorities. Where there is a conflict between a requirement of this Guideline and such other regulations affecting the design, construction and operation of the project, the building regulations shall take precedence.

3. Revision Log

Revision	Description	Date Effective
1.1	Issued for Implementation	1st June 2013
1.2	Revised version of Implementation	1st June 2014
2.0	Revised version of Implementation	1st June 2015
3.0	Revised version of Implementation	1st October 2015
3.1	Revised version of Implementation	15th March 2018
3.2	Revised version of Implementation	15th February 2021
3.2	Revised version of Implementation	1st September 2021
3.3	Revised version of Implementation	June 2023
3.3	Revised version of Implementation	January 2024
3.3	Revised version of Implementation	March 2024
3.3	Revised version of Implementation	April 2024
4.0	Revised version of Implementation	February 2026

4. GreenRE Assessment Stages

The GreenRE Existing Non-Residential Building certification process is as follows:



Application

- Submittal of application with relevant supporting documents for certification upon strategic inception of infrastructure project.



Pre-Assessment

- A pre-assessment can be conducted (optional) to give the project team a better understanding of the criteria and evaluation of the certification level sought. This should be performed upon selection of suitable design option to allow teams to identify and maximise opportunities at the earliest stages of the project.



Actual Assessment

- Actual assessment to be conducted once the design and documentary evidences (e.g. approved plan) are ready. After the actual assessment, our assessors will review the documents submitted.
- Assessment process includes design and documentary reviews to verify if the building project meets:
  i. The intents of the criteria
  ii. The prerequisite requirement for GreenRE Bronze, Silver, Gold and Platinum rating where applicable.
- Provisional Certificate will be issued upon completion of this stage.



Site Verification

- Site verification to be conducted upon project completion.
- Final Certificate will be issued upon completion of this stage.

5. GreenRE Existing Non-Residential Building Rating System

OVERVIEW:

The Existing Non-Residential Building Rating System is applicable to buildings that meet the following criteria:

- The building has been in operation for more than one (1) year; and
- The building maintains an occupancy rate of at least 60% (Energy Efficiency Compliance Option 2).

These conditions ensure that sufficient operational data is available to support a meaningful assessment of the building's performance under the rating system.

GreenRE assessment criteria consist of six (6) environmental impact categories namely:

1. **Part 1** – Energy Efficiency: This category focuses on the approach that can be used in the building design and system selection to optimise the energy efficiency of buildings.

2. **Part 2** – Water Efficiency: This category focuses on the selection of fittings and strategies enabling water use efficiency during construction and building operation.

3. **Part 3** – Sustainable Operation & Management: This category focuses on the sustainability of operation and management that would reduce the environmental impacts upon building operation.

4. **Part 4** – Indoor Environmental Quality: This category focuses on the design strategies that would enhance the indoor environmental quality which include air quality, thermal comfort, acoustic control and daylighting.

5. **Part 5** – Other Green Features: This category focuses on the adoption of green practices and new technologies that are innovative and have potential environmental benefits.

6. **Part 6** – Carbon Emission of Development: This category focuses on the use of carbon calculator to calculate the carbon emission of the development.

These environmental impact categories are broadly classified into two main groups: (I) Energy-Related Requirements and (II) Other Green Requirements.

Energy Related Requirements consist of Part 1- Energy Efficiency where credits are allocated for the various energy efficient designs, practices and features used. A minimum of 30 credits must be obtained from this group to be eligible for certification. The number of credits achievable for this group is capped at 50 credits (exclude 20 bonus credits that are obtainable under ENRB 1-10 – Renewable Energy).

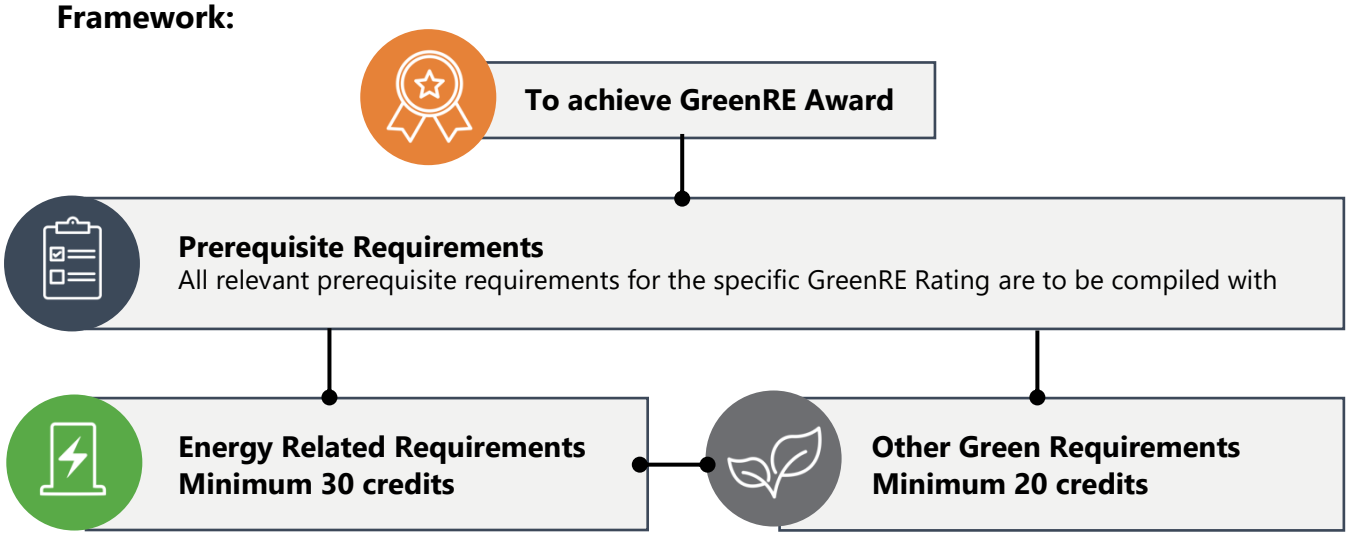
Other Green Requirements consist of Part 2 – Water Efficiency; Part 3 – Sustainable Operation & Management; Part 4 – Indoor Environmental Quality; Part 5 – Other Green Features and Part 6: Carbon Emission of Development. Credits are allocated for the water efficient features, environmentally friendly design practices, innovative green features used and carbon emission of development. A minimum of 20 credits must be obtained from this group to be eligible for certification. The number of credits achievable for this group is also capped at 50 credits.

The maximum GreenRE score achievable for a project is capped at 100 credits and this does not include 20 bonus credits that are obtainable under Energy Related Requirements if a project uses renewable energy sources.

Under the non-residential building criteria, the environmental impact category Part 1 – Energy Efficiency applies to both air-conditioned and non-air-conditioned spaces. Where there is a combination of air-conditioned and non-air-conditioned spaces, the credits allocated are to be prorated in accordance with the respective floor areas.

For simplicity, credits applicable to air-conditioned areas are accounted only if the aggregate air-conditioned areas exceed 1000 m^2. Similarly, credits applicable to non-air-conditioned areas are accounted only if the aggregate non-air-conditioned areas are more than 10% of the total floor areas excluding carparks and common areas.

Framework:



Elective Requirement for Energy Improvement (Combination of the following items to meet 30 credits)

PART 1 - ENERGY EFFICIENCY

ENRB 1-1 Thermal Performance of Building Envelope -OTTV
ENRB 1-2 Air-Conditioning System
ENRB 1-3 Natural Ventilation /Mechanical Ventilation
ENRB 1-4 Artificial Lighting
ENRB 1-5 Ventilation in Carparks
ENRB 1-6 Ventilation in Common Areas
ENRB 1-7 Lifts and Escalators
ENRB 1-8 Energy Efficient Practices & Features
ENRB 1-9 Energy Policy & Management
ENRB 1-10 Renewable Energy

Elective Requirement for Other Areas (Combination of the following items to meet 20 credits)

PART 2 - WATER EFFICIENCY

ENRB 2-1 Water Usage and Leak Detection
ENRB 2-2 Water Efficient Fittings
ENRB 2-3 Alternative Water Sources
ENRB 2-4 Water Efficiency Improvement Plans
ENRB 2-5 Irrigation System and Landscaping
ENRB 2-6 Water Consumption of Cooling Towers

PART 3 – ENVIRONMENTAL PROTECTION

ENRB 3-1 Building Operation & Maintenance
ENRB 3-2 Post Occupancy Evaluation
ENRB 3-3 Waste Management
ENRB 3-4 Sustainable Products
ENRB 3-5 Greenery Provision
ENRB 3-6 Environmental Protection
ENRB 3-7 Green Transport

PART 4 - INDOOR ENVIRONMENTAL QUALITY

ENRB 4-1 Indoor Air Quality Performance
ENRB 4-2 Indoor Air Pollutants
ENRB 4-3 Lighting Quality
ENRB 4-4 Thermal Comfort
ENRB 4-5 Noise Level

PART 5 – OTHER GREEN FEATURES

ENRB 5-1 Green Features & Innovations

PART 6 – CARBON EMISSION DEVELOPMENT

ENRB 6-1 Carbon Emission of Development

CREDIT ALLOCATION:

	Category	Credits Allocation
(I) Energy Related Requirements		
Minimum 30 credits	**Part 1: Energy Efficiency**	
	ENRB 1-1 Thermal Performance of Building Envelope-OTTV	5
	ENRB 1-2 Air-Conditioning System (applicable to air-conditioned areas)	33
	ENRB 1-3 Natural Ventilation /Mechanical Ventilation (Applicable to non-air-conditioned areas excluding carparks and common area)	32
	ENRB 1-4 Artificial Lighting	13
	ENRB 1-5 Ventilation in Carparks	4
	ENRB 1-6 Ventilation in Common Areas	5
	ENRB 1-7 Lifts & Escalators	2
	ENRB 1-8 Energy Efficient Practices & Features	12
	ENRB 1-9 Energy Policy & Management	1
	ENRB 1-10 Renewable Energy	20
	Category Score for Part 1 – Energy Efficiency	**95**
(II) Other Green Requirements		
Minimum 20 credits	**Part 2: Water Efficiency**	
	ENRB 2-1 Water Usage and Leak Detection System	4
	ENRB 2-2 Water Efficient Fittings	12
	ENRB 2-3 Alternative Water Sources	3
	ENRB 2-4 Water Efficiency Improvement Plans	1
	ENRB 2-5 Irrigation System and Landscaping	3
	ENRB 2-6 Water Consumption of Cooling Towers	2
	Category Score for Part 2 – Water Efficiency	**25**
	Part 3: Sustainable Operation & Management	
	ENRB 3-1 Building Operation & Maintenance	5
	ENRB 3-2 Post Occupancy Evaluation	3
	ENRB 3-3 Waste Management	7
	ENRB 3-4 Sustainable Products	8
	ENRB 3-5 Greenery Provision	8
	ENRB 3-6 Environmental Protection	3
	ENRB 3-7 Green Transport	4
	Category Score for Part 3 – Sustainable Operation & Management	**38**
	Part 4: Indoor Environmental Quality	
	ENRB 4-1 Indoor Air Quality Performance	8
	ENRB 4-2 Indoor Air Pollutants	2
	ENRB 4-3 Lighting Quality	5
	ENRB 4-4 Thermal Comfort	2
	ENRB 4-5 Internal Noise Level	1
	Category Score for Part 4: Indoor Environmental Quality	**18**
	Part 5: Sustainable Practices and Green Innovation	
	ENRB 5-1 Green Features & Innovations	10
	Category Score for Part 5: Other Green Features	**10**
	Part 6: Carbon Emission of Development	
	ENRB 6-1 Carbon Emission of Development	2
	Category Score for Part 6: Carbon Emission of Development	**2**
	Category Score for Part 6 – Carbon Emission of Development	**93**
	GreenRE Existing Non-Residential Building Score:	**188**

6. GreenRE Existing Non-Residential Building Rating System Scoring

GreenRE Platinum	GreenRE Gold	GreenRE Silver	GreenRE Bronze
			
Score: 91 and above	Score: 86 to ≤ 90	Score: 76 to ≤ 85	Score: 50 to ≤ 75

7. GreenRE Existing Non-Residential Building Rating System Criteria

PREREQUISITE

Part 1 – Energy Efficiency

ENERGY EFFICIENCY

GreenRE Bronze	GreenRE Silver	GreenRE Gold	GreenRE Platinum
Minimum credits: 30 credits	Minimum credits: 35 credits	Minimum credits: 40 credits	Minimum credits: 45 credits

- Building Energy Intensity (BEI) calculation, proper submetering is essential. Projects must ensure that energy consumption is accurately tracked through dedicated submeters.

ENERGY EFFICIENCY COMPLIANCE

Projects shall demonstrate the stipulated performance through either option listed below;

Option 1: Minimum System Efficiency (Fixed Metric)

Minimum Design System Efficiency/Operating System Efficiency (DSE/OSE)

- For buildings using Water-Cooled Chilled Water Plant

GreenRE Rating	Building Cooling Load (RT)	
	< 500	≥ 500
	Efficiency (kW/RT)	
Bronze	0.85	0.75
Silver	0.80	0.70
Gold	0.75	0.68
Platinum	0.70	0.65

- For buildings using Air-Cooled Chilled Water Plant or Unitary Air-Conditioner

GreenRE Rating	Building Cooling Load (RT)	
	< 500	≥ 500
	Efficiency (kW/RT)	
Bronze	1.1	1.0
Silver	1.0	1.0
Gold	0.85	Case by case (i)
Platinum	0.78	

For building with building cooling load of more than 500RT, the use of air cooled central chilled water plant or other unitary air-conditioners are not encouraged for Gold and Platinum ratings. In general, the system efficiency of the air cooled central chilled-water plant and other unitary air-conditioners are to be comparable with the stipulated efficiency for water-cooled central chilled-water plant. Buildings that are designed with air cooled systems and for higher GreenRE rating will be assessed on a case-by-case basis.

Note: The performance of the overall air-conditioning system for the building is based on the Operating System Efficiency (OSE) of the system during normal building operating hours as defined below:

Office Building
Monday to Friday: 9am to 6pm

Retail Mall:
Monday to Sunday: 10am to 9pm

Institutional:
Monday to Friday: 9am to 5pm

Hotel and Hospital:
24-hour

Industrial and Other Building Types:
To be determined based on the operating hours

Chiller Plant M&V Instrument (Gold & Platinum)

For **Gold & Platinum**, the project also needs to provide the permanent measuring instruments for monitoring of water-cooled chilled-water system and air-cooled chilled water system operating system efficiency. The installed instrumentation shall have the capability to calculate resultant plant operating system efficiency (i.e., kW/RT) within 5% of its true value and in accordance with ASHRAE Guide 22 and AHRI 550/590. Heat balance test for water-cooled chilled water system is required for verification of the accuracy of the Measurement and Verification (M&V) instrumentation.

Option 2 – Building Energy Intensity (BEI) Benchmarking

Total Building annual energy consumption over the gross floor area of the building (kWh/m^2/yr). Based on:

- Energy Calculation and measured data (Retrofit)
- Measurement – In operation

The building shall demonstrate compliance to the Building Energy Intensity (BEI) stated in the table below through 12-months measured data with a requirement of minimum occupancy of 60% for the period of measurement (projection to 80% occupancy of energy consumption necessary for lower occupancy <20% based on prescribed BEI formula):

BEI is derived using the following equation:

BEI = [(TBEC – CPEC) / (GFA excluding carpark - GLA x FVR) x (NF/OH)

Where:
TBEC = Total building energy consumption excluding renewable energy replacement (kWh/year)
CPEC = Car Park Energy Consumption in (kWh/year)
GFA = Gross Floor Area (exclude car park area) (m^2)
GLA = Gross Lettable Area (m^2)
FVR = Floor Vacancy Rate (NLA) (m^2)
NF = Normalizing factor based on a typical weekly operating hour that is 52 Hrs/week [only for office category]
OH = Weighted weekly operating hours (hrs/week) [only for office category]

Building Type	Bronze (kWh/m²/year)	Silver (kWh/m²/year)	Gold (kWh/m²/year)	Platinum (kWh/m²/year)
Office Buildings (Small) (GFA < 15,000sqm)	205	180	135	120
Office Buildings (Large) (GFA > 15,000sqm)	210	190	155	140
Retail Mall	360	315	230	210
School, University and College	170	145	110	200
School, University and College (MOE)	60	50	40	35
Community Centres	210	190	150	125
Civic Buildings	120	105	80	70
Cultural Institutions	240	210	180	140
Sport and Recreations Centres	135	120	110	80

Table 1: Building Energy Intensity (BEI) Benchmarking

Building Type	Bronze (kWh/m²/year)	Silver (kWh/m²/year)	Gold (kWh/m²/year)	Platinum (kWh/m²/year)
Office Buildings (Small) (GFA < 15,000sqm)	150	125	100	90
Office Buildings (Large) (GFA > 15,000sqm)	130	120	90	80
Retail Mall	230	200	160	140
Community Centres	150	130	100	90
Civic Buildings	70	60	50	45
Cultural Institutions	160	135	115	100
Sport and Recreations Centres	100	85	70	65

Table 2: Building Energy Intensity (BEI) Benchmarking - District Cooling System (DCS)

NATURAL VENTILATION (only applicable to occupied areas, excluding circulation, plant rooms and transit areas)

Prerequisite requirement for Platinum - At least 75% of natural ventilated areas with effective cross ventilation with North and South facing window opening.

Part 3 – Sustainable Operations & Management

Provision of Building User Guide and Sustainable Operation Management Guide

Part 4 – Indoor Environmental Quality

A formal IAQ management plan must be established and implemented to ensure a healthy indoor environment for all occupants. The policy should outline procedures for maintaining acceptable indoor air quality, including ventilation, pollutant source control, and regular maintenance of HVAC systems. The policy should contain at least the following:

- **Ventilation Management**: Ensure adequate ventilation rates in accordance with recognized standards (e.g., ASHRAE 62.1 or local equivalents) to dilute indoor pollutants and maintain fresh air supply.
- **Pollutant Source Control:** Identify and minimize sources of indoor air contaminants, including volatile organic compounds (VOCs), particulate matter, and biological pollutants. This includes the selection of low-emission materials and proper storage of chemicals.
- **HVAC System Maintenance:** Establish a routine inspection and maintenance schedule for HVAC systems to ensure optimal performance, cleanliness, and efficiency. Filters should be replaced regularly, and air ducts cleaned as needed.
- **Monitoring and Response:** Implement procedures for periodic IAQ monitoring and establish a response procedure for occupant complaints or IAQ-related incidents.
- **Occupant Awareness:** Promote awareness and education among building users regarding IAQ practices, including the importance of reporting issues and maintaining cleanliness.



Part 1

Energy Efficiency

Part 1 – Energy Efficiency

ENRB 1-1 THERMAL PERFORMANCE OF BUILDING ENVELOPE - OTTV	GREENRE CREDITS
Enhance the overall thermal performance of building envelope to minimize heat gain thus reducing the overall cooling load requirement. Baseline: Maximum permissible OTTV = 50 W/m^2	0.5 credits for every reduction of 1 W/m^2 in OTTV from the baseline of 50 W/m^2. Credits scored = 0.5 x (50 - OTTV) (Up to 5 credits)

ENRB 1-2 AIR-CONDITIONING SYSTEM	GREENRE CREDITS
Option 1 – Fixed Metrics Applicable to Air-conditioned Building Areas (with an aggregate air-conditioned area > 1000m^2) Encourage the use of better efficiency air-conditioned equipment to minimize the energy consumption. (System efficiency in kW/RT)	
a. Water-Cooled Chilled-Water Plant: • Water-Cooled Chiller • Chilled water pump • Condenser water pump • Cooling tower	a. Water-Cooled Chilled-Water Plant: Building cooling load < 500 RT 14 credits for achieving plant efficiency of 0.85 kW/RT 0.3 credit for every percentage improvement in the chiller plant efficiency better than 0.85 kW/RT Credit scored = 0.3 x (% improvement)

Baseline	Building Cooling Load	
	<500 RT	≥500RT
Prerequisite Requirements Minimum Design System Efficiency (DSE) for central chilled-water plant	0.85 kW/RT	0.75 kW/RT

OR

b. Air Cooled Chilled-Water Plant / Unitary Air-Conditioners:

Air cooled Chilled-Water Plant:
- Air-Cooled Chiller
- Chilled Water Pump

Unitary Air-Conditioners:
- Variable Refrigerant Flow (VRF) System
- Water-Cooled Package Unit
- Single-Split Unit
- Multi-Split Unit

Baseline	Building Cooling Load	
	< 500 RT	≥ 500 RT
Prerequisite Requirements Minimum system efficiency of air-cooled chilled water plant or unitary conditioners	1.1 kW/RT	1.0 kW/RT

Note (1): When there is a combination of centralised air-conditioned system with unitary air-conditioned system, the computation for the credits scored will be pro-rated based on the air-conditioning system aggregate capacity.

c. Air Distribution system:

- Air Handling units (AHUs)
- Fan Coil Units (FCUs)

Fan System Input Power
Baseline: ASHRAE 90.1:2010 Clause 6.5.3.1 and as prescribed below;

Building cooling load ≥ 500 RT

14 credits for achieving plant efficiency of

0.75 kW/RT

0.35 credit for every percentage improvement in the chiller plant efficiency better than 0.75 kW/RT

Credit scored = 0.35 x (% improvement)

(up to 20 credits)

OR

b. Air Cooled Chilled-Water Plant / Unitary Air-Conditioners:

Building cooling load < 500 RT

14 credits for achieving plant efficiency of

1.1 kW/RT

0.2 credit for every percentage improvement in the chiller plant efficiency better than 1.1 kW/RT

Credit scored = 0.2 x (% improvement)

Building cooling load ≥ 500 RT

14 credits for achieving plant efficiency of

1.0 kW/RT

0.25 credit for every percentage improvement in the chiller plant efficiency better than 1.0 kW/RT

Credit scored = 0.25 x (% improvement)

(up to 20 credits)

c. Air Distribution system:

0.15 credits for every percentage improvement in the air distribution system efficiency over the baseline

Credit scored = 0.15 x (% improvement)

(up to 8 credits)

Baseline Air Distribution System Type	Allowable Fan System Input Power	
	(kW/m³/s)	(W/CMH)
AHUs / FCUs ≥ 4kW (Constant Volume)	1.5	0.42
AHUs ≥ 4kW (Variable Volume)	2.1	0.58
Fan systems with nameplate motor power < 4kW	0.6	0.17

Note (2): For buildings using district cooling system, there is no need to compute the plant efficiency under Part 1-2 (a) and (b). The credits obtained will be pro-rated based on the air distribution system efficiency under Part 1-2(c).

d. Prerequisite requirements: Provision of permanent measuring instruments for monitoring of water-cooled chilled water plant and air-cooled chilled water plant efficiency. The installed instrumentation shall have the capability to calculate resultant plant efficiency (i.e. kW/RT) within 5% of its true value and in accordance with ASHRAE Guide 22 and AHRI 550/590. The following instrumentation and installation are also required to be complied:

- Location and installation of the measuring devices to meet the manufacturer's and engineer's recommendation.
- Data acquisition system to have a minimum resolution of 16 bit.
- All data logging with capability to trend at 1 minute sampling time interval for 24 hours a day, 7 days a week.
- Dedicated digital power meters shall be provided for the each of equipment: chiller(s), chilled water pump(s), condenser water pump(s) and cooling tower(s).

Applicable only to buildings with provision of water-cooled chilled water plants

2 credits

• Flow meters to be provided for chilled-water and condenser water loop (return and supply) and shall be of ultrasonic / full bore magnetic type or equivalent. • Temperature sensors are to be provided for chilled water and condenser water loop (return and supply) and shall have an end-to-end measurement uncertainty not exceeding ± 0.05°C over entire measurement or calibration range. All thermo-wells shall be installed in a manner that ensures that the sensors can be in direct contact with fluid flow. Provisions shall be made for each temperature measurement location to have two spare thermo-wells located at both side of the temperature sensor for verification of measurement accuracy.	
e. Verification of central water cooled chilled-water plant instrumentation: Heat Balance – substantiating test for water cooled chilled-water plant to be computed in accordance with AHRI 550/590. The operating system efficiency and heat balance to be submitted to GreenRE upon commissioning.	1 credit
f. Provision of variable speed controls for chiller plant equipment such as chilled-water pumps and cooling tower fans to ensure better part-load plant efficiency.	1 credit
g. Sensors or similar automatic control devices are used to regulate outdoor air flow rate to maintain the concentration of carbon dioxide. Indoor carbon dioxide acceptable range ≤700 ppm above outdoor concentration.	1 credit
Option 2 – BEI Benchmarking **<u>Air-conditioning System</u>** a. Projects with air conditioning system Total Building annual energy consumption over the gross floor area of the building (kWh/m^2/yr). Based on: • Energy Calculation and measured data (Retrofit) • Measurement – In operation	14 credits for achieving BEI per table shown.

The project shall demonstrate the Building Energy Intensity (BEI) and show compliance to the table below (minimum occupancy >80%)-(projection of energy consumption necessary for lower occupancy based on prescribed BEI formula):

BEI = [(TBEC – CPEC) / GFA excluding carpark – GLA x FVR) x (NF/OH)

Where:

TBEC = Total building energy consumption without the renewable energy replacement (kWh/year)

CPEC = Car Park Energy Consumption in (kWh/year)

GFA = Gross Floor Area (exclude car park area) (m^2)

GLA = Gross Lettable Area (m^2)

FVR = Floor Vacancy Rate (NLA) (m^2)

NF = Normalizing factor based on a typical weekly operating hour that is 52 Hrs/week [only for office category]

OH = Weighted weekly operating hours (hrs/week) [only for office category]

Building Type	Bronze	Silver	Gold	Platinum
Office Buildings (Small) (GFA < 15,000sqm)	205	180	135	120
Office Buildings (Large) (GFA > 15,000sqm)	210	190	155	140
Retail Mall	360	315	230	210
School, University and College	170	145	110	100
School, University and College (MOE)	60	50	40	35
Community Centres	210	190	150	125
Civic Buildings	120	105	80	70
Cultural Institutions	240	210	180	140
Sport and Recreations Centres	135	120	110	80

Table 1: Building Energy Intensity (BEI) Benchmarking (kwh/m2/yr)

b. Water-Cooled Chilled-Water Plant:
- Water-Cooled Chiller
- Chilled water pump
- Condenser water pump
- Cooling tower

Baseline	Building Cooling Load	
	< 500 RT	≥500 RT
Prerequisite Requirements Minimum system efficiency of chilled-water plant	0.85 kW/RT	0.75 kW/RT

OR

b. Water-Cooled Chilled-Water Plant

Building cooling load < 500 RT

0.3 credit for every percentage improvement in the chiller plant efficiency better than 0.85 kW/RT

Credits scored = 0.3 x (%improvement)

Building cooling load ≥ 500 RT

0.35 credit for every percentage improvement in the chiller plant efficiency better than 0.75 kW/RT

Credit scored = 0.35 x % improvement)

(up to 8 credits)

OR

c. Air Cooled Chilled-Water Plant / Unitary Air-Conditioners:

Air cooled Chilled-Water Plant:
- Air-Cooled Chiller
- Chilled Water Pump

Unitary Air-Conditioners:
- Variable Refrigerant Flow (VRF) System
- Water-Cooled Package Unit
- Single-Split Unit
- Multi-Split Unit

Baseline	Building Cooling Load	
	< 500 RT	≥ 500 RT
Prerequisite Requirements Minimum system efficiency of air-cooled chilled water plant or unitary conditioners	1.1 kW/RT	1.0 kW/RT

Note (1): When there is a combination of centralised air-conditioned system with unitary air-conditioned system, the computation for the credit scored will be pro-rated based on the air-conditioning system aggregate capacity.

c. Air Cooled Chilled-Water Plant / Unitary Air-Conditioners:

Building cooling load < 500 RT

0.2 credit for every percentage improvement in the chiller plant efficiency better than 1.1 kW/RT

Credit scored = 0.2 x (% improvement)

Building cooling load ≥ 500 RT

0.25 credit for every percentage improvement in the chiller plant efficiency better than 1.0 kW/RT

Credit scored = 0.25 x (% improvement)

(up to 8 credits)

d. Provision of permanent measuring instruments for monitoring of water-cooled chilled water plant and air-cooled chilled water plant efficiency. The installed instrumentation shall have the capability to calculate resultant plant efficiency (i.e., kW/RT) within 5% of its true value and in accordance with ASHRAE Guide 22 and AHRI 550/590. The following instrumentation and installation are also required to be complied: • Location and installation of the measuring devices to meet the manufacturer's recommendation. • Data acquisition system to have a minimum resolution of 16 bit. • All data logging with capability to trend at 1minute sampling time interval. • Dedicated digital power meters shall be provided for the following groups of equipment: chiller(s), chilled water pump(s), condenser water pump(s) and cooling tower(s). • Flow meters to be provided for chilled-water and condenser water loop and shall be of ultrasonic / full bore magnetic type or equivalent. • Temperature sensors are to be provided for chilled water and condenser water loop and shall have an end-to-end measurement uncertainty not exceeding ± 0.05°C over entire measurement or calibration range. All thermo-wells shall be installed in a manner that ensures that the sensors can be in direct contact with fluid flow. Provisions shall be made for each temperature measurement location to have two spare thermo-wells located at both side of the temperature sensor for verification of measurement accuracy.	Applicable only to buildings with provision of water-cooled chilled water plants 2 credits
e. Verification of central water cooled chilled-water plant instrumentation: Heat Balance - substantiating test for water cooled chilled-water plant to be computed in accordance with AHRI 550/590. The operating system efficiency and heat balance to be submitted to GreenRE upon commissioning.	1 credit
f. Provision of variable speed controls for chiller plant equipment such as chilled-water pumps and cooling tower fans to ensure better part-load plant efficiency	1 credit

g. Sensors or similar automatic control devices are used to regulate outdoor air flow rate to maintain the concentration of carbon dioxide. Indoor carbon dioxide acceptable range ≤700 ppm above outdoor concentration.

1 credit

District Cooling System

a. Projects with air conditioning system

Total Building annual energy consumption over the gross floor area of the building (kWh/m^2/yr). Based on:

- Energy Calculation and measured data (Retrofit)
- Measurement – In operation

The project shall demonstrate the Building Energy Intensity (BEI) and show compliance to the table below (minimum occupancy >80%)

Building Type	Bronze	Silver	Gold	Platinum
Office Buildings (Large) (GFA < 15,000sqm)	150	125	100	90
Office Buildings (Small) (GFA > 15,000sqm)	130	120	90	80
Retail Mall	230	200	160	140
Community Centres	150	130	100	90
Civic Buildings	70	60	50	45
Cultural Institutions	160	135	115	100
Sport and Recreations Centres	100	85	70	65

Table 2: Building Energy Intensity (BEI) Benchmarking - District Cooling System (DCS)

14 credits for achieving BEI per table shown.

b. Air Distribution system:
- Air Handling units (AHUs)
- Fan Coil Units (FCUs)

Fan System Input Power

Baseline: ASHRAE 90.1:2010 Clause 6.5.3.1 and as prescribed below;

Baseline Air Distribution System Type	Allowable Fan System Input Power	
	(kW/m³/s)	(W/CMH)
AHUs / FCUs ≥ 4kW (Constant Volume)	1.5	0.42
AHUs ≥ 4kW (Variable Volume)	2.1	0.58
Fan systems with nameplate motor power < 4kW	0.6	0.17

b. Air Distribution system:

0.15 credits for every percentage improvement in the air distribution system efficiency over the baseline

Credits scored = 0.15 x (% improvement)

(up to 8 credits)

ENRB 1-3 NATURAL VENTILATION / MECHANICAL VENTILATION

GREENRE CREDITS

Applicable to Non-Air-Conditioned Building Areas (with an aggregate non-air-conditioned areas > 10% of total floor area excluding carparks and common areas)

a. Natural Ventilation

(only applicable to occupied areas, excluding circulation, plant rooms and transit areas)

Encourage building that facilitates good natural ventilation. Proper design of building layout that utilises prevailing wind conditions to achieve adequate cross ventilation.

Note (3): credit can be attempted if window openings facing **predominant wind directions** and cross ventilation. Local wind rose to be submitted to justify predominant wind directions.

20 based credits will be awarded for use of natural ventilation

1.2 credits for every 10% of NV areas with window openings facing north and south directions and cross ventilation

(Up to 32 credits)

b. Mechanical Ventilation

Encourage energy efficient mechanical ventilation system and analysis

Baseline: Fan power limitation in mechanical ventilation systems:

Allowable nameplate motor power	
Nameplate motor power > 4kW	Nameplate motor power < 4kW
0.32 W/CMH	0.17 W/CMH

Note (3): When there is a combination of naturally ventilated and mechanical ventilated spaces, the credits scored will only be based on the predominant ventilation modes of normally occupied spaces.

0.6 credit for every subsequent 1% improvement from the baseline

(Up to 32 credits)

ENRB 1-4 ARTIFICIAL LIGHTING	GREENRE CREDITS
Encourage the use of better efficient lighting to minimise energy consumption from lighting usage while maintaining proper lighting level. Baseline: Luminance level stated in MS 1525:2019-Energy Efficient and use of renewable energy for non-residential building – Code of Practice	0.25 credit for every percentage improvement in the lighting power budget Credits scored = 0.25 x (% improvement) (Up to 13 credits) Excluding tenant lighting provision (Up to 5 credits)

ENRB 1-5 VENTILATION IN CARPARKS	GREENRE CREDITS
Encourage the use energy efficient design and control of ventilation systems on carparks. a. Carparks designed with natural ventilation. b. CO sensors are used to regulate the demand for mechanical ventilation (MV). Note (4): When there is a combination of different ventilation mode adopted for car park design, the credits scored under this requirement will be prorated accordingly.	Naturally ventilated carparks – 4 credits Credits scored based on the mode of mechanical ventilation provided: Fume extract- 2.5 credit MV with or without supply – 2 credits (Up to 4 credits)

ENRB 1-6 VENTILATION IN COMMON AREAS	GREENRE CREDITS
Encourage the use of energy efficient design and control of ventilation systems in the following common areas: • Toilets • Staircases (Including BOMBA Staircase) • Lift Lobbies • Corridors • Atriums	Extent of Coverage: At least 90% of each applicable area Credits scored based on the mode of ventilation provided in the applicable areas. Natural Vent. – 1.5 credits for each area Mechanical Vent. – 0.5 credit for each area (Up to 5 credits)

ENRB 1-7 LIFTS AND ESCALATORS	GREENRE CREDITS
a. Encourage the use of lift with energy efficient features such as sleep mode / regenerative features / motorless lift.	Extent of Coverage: All lifts and/or escalators 1 credit
b. Escalators with energy efficient features such as motion sensors.	1 credit

ENRB 1-8 ENERGY EFFICIENT PRACTICES & FEATURES	GREENRE CREDITS
Encourage the use of energy efficient practices and features which are innovative and have positive environmental impact	
a. Computation of energy consumption based on design load in the form of Building Energy Intensity (BEI)	1 credit
b. Use of energy efficiency product that are certified by approved local certification body	0.5 credit for each equipment type (Up to 2 credits)
c. Use of energy efficient features: Examples: • Re-generative lift • Heat recovery system • Motion sensors • Sun pipes • Light shelves • Photocell sensor to maximise the use of daylight • Heat pumps, etc.	3 credits for every 1% energy saving over the total building energy consumption per features (Up to 11 credits)

ENRB 1-9 ENERGY POLICY AND MANAGEMENT	GREENRE CREDITS
a. Energy policy, energy targets and regular review with top management's commitment as part of an environmental strategy	0.5 credit
b. To show intent, measures and implementation strategies of energy efficiency improvement plans to achieve energy target set over the next three years. Committed energy savings accrued from proposed measures should be quantified.	0.5 credit

ENRB 1-10 RENEWABLE ENERGY	GREENRE CREDITS
Encourage the use of renewable energy sources in buildings	5 credits for every 1% replacement of electricity (based on total electricity consumption) by renewable energy OR 3 credits for every 1% replacement of electricity (based on the total electricity consumption excluding tenant's usage) by renewable energy OR 3 credits for every 10% of roof area used for solar panels. Note: The credit scored for renewable energy provision shall not result in a double grade jump in GreenRE rating (i.e from GreenRE Bronze or Silver to Gold or Platinum) (Up to 20 credits)
Sub-Total (C):	Sum of GreenRE credits obtained from NRB 1-5 to 1-11
PART 1 – ENERGY EFFICIENCY CATEGORY SCORE:	(ENRB 1-2) X $\frac{\text{Air-Conditioned Building Floor Area}}{\text{Total Floor Area}}$ + (ENRB 1-3) X $\frac{\text{Non-Air-Conditioned Building Floor Area}}{\text{Total Floor Area}}$ + (ENRB 1-1, ENRB 1-4 to ENRB 1-10) Where: ENRB 1-2 = Total GreenRE Credits obtained under is ENRB 1-2 ENRB 1-3 = Sum of GreenRE Credits obtained under ENRB 1-3 ENRB 1-1, ENRB 1-4 to ENRB 1-10 = Sum of GreenRE Credits obtained under ENRB 1-1, ENRB 1-4 to ENRB 1-10



Part 2

Water Efficiency

Part 2 – Water Efficiency

ENRB 2-1 WATER USAGE AND LEAK DETECTION	GREENRE CREDITS
Provide sub-metering and leak detection system for better control and monitoring	
a. To monitor the water consumption on monthly basis	1 credit
b. Provision of sub-meters for major water uses (e.g., cooling tower, water features, irrigation, swimming pools, tenants' usage and the non-portable water replacement system)	1 credit
c. Provision of automated / smart metering for monitoring and leaking detection	2 credits

ENRB 2-2 WATER EFFICIENT FITTINGS	GREENRE CREDITS
Encourage the use of water efficient fittings under the Water Efficiency Products Labelling Scheme (WEPLS). • Basin taps and mixers • Shower taps and mixers or showerheads • Sink/bib taps and mixers • Urinals and Urinal Flush Valves • Dual flushing cisterns for WC • Other water fittings (e.g. Ablution taps and mixers)	

Weightage Based on Water Efficiency Products Labelling Scheme (WEPLS)		
Efficient*	Highly Efficient**	Most Efficient***
6	**9**	**12**

Credits can be scored based on the number and water efficiency rating of the fitting type used.
(Up to 12 credits)

ENRB 2-3 ALTERNATIVE WATER SOURCES	GREENRE CREDITS
Use of suitable systems that utilize alternative water sources for **non-potable uses**: irrigation, washing, water features, toilet flushing, etc (excluding cooling tower make up water) to reduce use of potable water. Alternative sources can include rainwater, greywater (for toilet flushing only), AHU condensate and recycled water from approved sources with the provision of submetering for monitoring during the operation.	Credits awarded based on % reduction in total potable water usage of the applicable uses > 50 % — 3 credits ≥ 10 % to 50 % — 2 credits < 10 % — 1 credit (Up to 3 credits)

ENRB 2-4 WATER EFFICIENCY IMPROVEMENT PLANS	GREENRE CREDITS
Targets to improve building water performance against own building water performance baseline should be set. To show intent, measures and implementation strategies of water efficiency improvement plans over the next three years. Committed water savings accrued from proposed measures should be quantified.	1 credit

ENRB 2-5 IRRIGATION SYSTEM AND LANDSCAPING	GREENRE CREDITS
Reduce potable water consumption for irrigation and landscaping.	
a. Use of non-potable water including rainwater for landscape irrigation with provision of submetering for monitoring during the operation.	1 credit
b. Use of automatic water efficient irrigation system with rain sensor, soil moisture sensor or equivalent control system.	Extent of Coverage: At least 50% of the landscape areas are served by the system 1 credit
c. Use of drought tolerant plants that require minimal irrigation.	Extent of Coverage: At least 80% of the landscape areas 1 credit

ENRB 2-6 WATER CONSUMPTION OF COOLING TOWER	GREENRE CREDITS
Reduce potable water consumption for cooling purpose.	
a. Use of cooling tower water treatment system which can achieve 6 or better cycles of concentration at acceptable water quality	1 credit
b. Use of recycled water from approved sources for cooling purpose	1 credit
PART 2 – WATER EFFICIENCY CATEGORY SCORE:	Sum of GreenRE credits obtained from ENRB 2-1 to 2-6



Part 3

Sustainable Operations & Management

Part 3 – Sustainable Operations & Management

ENRB 3-1 BUILDING OPERATION & MAINTENANCE	GREENRE CREDITS
a. The environmental policy reflects the sustainability goals set.	1 credit
b. Provision of a Building User Guide	1 credit
c. In-house building management team comprises one Certified GreenRE Accredited Professional (GREAP) / Green Mark Accredited Professional (GMAP)	1 credit
d. Project team comprises one Certified GreenRE Accredited Professional (GREAP) / Green Mark Accredited Professional (GMAP)	1 credit
e. The environmental management system of the building is ISO14000 or ISO 50001 certified.	1 credit

ENRB 3-2 POST OCCUPANCY EVALUATION	GREENRE CREDITS
a. Conduct post occupancy survey for occupant's satisfaction on energy and environmental performance. Required number of people surveyed shall be: • 10% of total occupancy and up to 100 maximums. • Minimum 5 people shall be surveyed if total occupancy is less than 50 people.	2 credits
b. List of corrective actions taken following the post occupancy evaluation, if any.	1 credit

ENRB 3-3 WASTE MANAGEMENT	GREENRE CREDITS
a. Provision of facilities or recycling bins for collection and storage of different recyclable waste such as paper, glass, plastic, food waste, etc.	2 credits
b. Promote and encourage waste minimization and recycling among occupants, tenants and visitors through various avenues	2 credits
c. Provide the proper storage area for the recyclable waste	1 credit
d. To quantify and monitor the recycling programme for continuous improvement.	2 credits

ENRB 3-4 SUSTAINABLE PRODUCTS	GREENRE CREDITS
Promote use of environmentally friendly products that are certified by approved local certification body and are applicable to non-structural and architectural related building components.	<table><tr><th>Extent of use of environmentally friendly product</th><th>Weightage for Credit Allocation</th></tr><tr><td>Low Impact</td><td>0.5</td></tr><tr><td>Medium impact</td><td>1</td></tr><tr><td>High Impact</td><td>2</td></tr></table> Credits scored will be based on the extent of use of environmentally friendly product. (Up to 8 credits)

ENRB 3-5 GREENERY PROVISION	GREENRE CREDITS
Encourage greater use of greenery to reduce heat island effect.	
a. Green Plot Ratio (GnPR) is calculated by considering the 3D volume covered by plants using the Leaf Area Index (LAI).	<table><tr><th>GnPR</th><th>Credits Allocation</th></tr><tr><td>1.0 to < 2.0</td><td>1</td></tr><tr><td>2.0 to < 3.0</td><td>2</td></tr><tr><td>3.0 to < 4.0</td><td>3</td></tr><tr><td>4.0 to < 5.0</td><td>4</td></tr><tr><td>5.0 to < 6.0</td><td>5</td></tr><tr><td>≥ 6.0</td><td>6</td></tr></table>
b. Restoration of trees on site, conserving or relocating of existing trees on site.	1 credit
c. Provision of compost bins to recycle organic waste to meet at least 30% of landscape fertilizer needs.	1 credit

ENRB 3-6 ENVIRONMENTAL PROTECTION	GREENRE CREDITS
a. Green procurement policy – Adoption of sustainable and environmental-friendly procurement and purchasing policy in the operation and maintenance of the building.	1 credit
b. Reduce the potential damage to the ozone layer and the increase in global warming through the release of ozone depleting substances and greenhouse gases.	1 credit
• Refrigerants with ozone depletion potential (ODP) of zero or with global warming potential (GWP) of less than 100.	1 credit
• Use of refrigerant leak detection system at critical areas of plant rooms containing chillers and other equipment with refrigerants.	1 credit

ENRB 3-7 GREEN TRANSPORT	GREENRE CREDITS
Promote the use of public transport or bicycles to reduce pollution from individual car use with the following provision:	
a. Good access (<800m walking distance) to public transport networks such as MRT/LRT stations or bus stops.	1 credit
b. Provision of covered walkway to facilitate connectivity and the use of public transport.	1 credit
c. Provision of hybrid/electric vehicle charging stations and priority parking lots within the development.	Extent of coverage: Minimum 1 number priority parking bays for every 100 carpark lots. EV chargers – 1 for every 200 parking bays. (Cap at 3) (1 credit)
d. Provision of covered / sheltered bicycles parking lots with adequate shower and changing facilities.	Extent of Coverage: Bicycles parking lot: Minimum 10 number and maximum 50 numbers of bicycle parking lot. Shower Facilities: Minimum 1 number for every 100 regular occupant and additional 1 for every 150 occupants. (Cap at 7) (1 credit)
PART 3–SUSTAINABLE OPERATION & MANAGEMENT CATEGORY SCORE:	Sum of GreenRE credits obtained from ENRB 3-1 to 3-7



Part 4

Indoor Environmental Quality

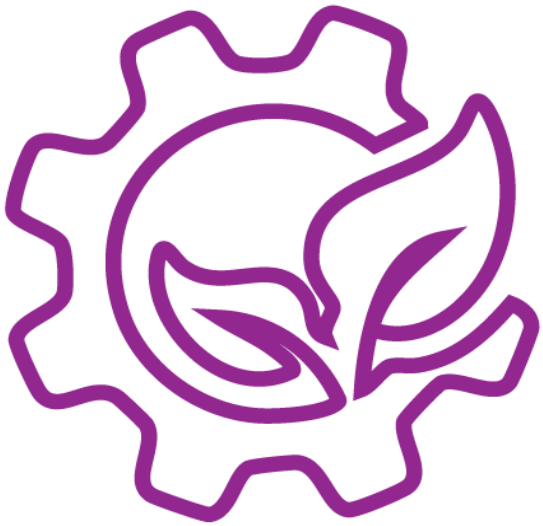
Part 4 – Indoor Environmental Quality

ENRB 4-1 INDOOR AIR QUALITY PERFORMANCE	GREENRE CREDITS
To promote a healthy indoor environment.	
a. *Prerequisite Requirements:* To conduct full IAQ audit once in three years that complies with Code of Practice on Indoor Air Quality, Department of Occupational Safety and Health, Ministry of Human Resources Malaysia (2005).	4 credits
b. Implement effective IAQ management plan to ensure building ventilation systems are frequently maintained to ensure clean delivery of air.	1 credit
c. Use of high efficiency air filter (at least MERV 8) in AHU to reduce indoor contaminants and provide good protection for cooling coil and reducing frequency or eliminating duct cleaning	1 credit
d. Room Temperature display (at least 1 unit per floor)	1 credit
e. Additional carbon dioxide sensor displays (at least 1 unit per floor)	1 credit

ENRB 4-2 INDOOR AIR POLLUTANTS	GREENRE CREDITS
Minimise airborne contaminants, mainly from inside sources to promote a healthy indoor environment.	
a. Use of low volatile organic compounds (VOC) paints certified under local/international certification body.	1 credit
b. Use adhesives certified under local/international certification body for composite wood products.	1 credit

<table>
<tr><th>NRB 4-3 LIGHTING QUALITY</th><th>GREENRE CREDITS</th></tr>
<tr><td>To encourage good workplace lighting quality to promote productivity and occupant's comfort</td><td></td></tr>
<tr><td>a. Lighting level to comply with MS1525:2019</td><td>1 credit</td></tr>
<tr><td>b. Controllability of lighting system</td><td>At least 90% of occupants are able to adjust lighting to suit their task needs and preference
<table><tr><td>Controlled by light switches</td><td>1 credit</td></tr><tr><td>Controlled by task lights</td><td>2 credits</td></tr></table>(Up to 2 credits)</td></tr>
<tr><td>c. High frequency ballast OR use of driver with output frequency < 200Hz and < 30% flicker for LED lighting.</td><td>All applicable areas in the entire building that are served by fluorescent / LED lighting.
<table><tr><td>20% to < 40%</td><td>0.5 credit</td></tr><tr><td>40% to < 60%</td><td>1 credit</td></tr><tr><td>60% to < 80%</td><td>1.5 credits</td></tr><tr><td>80% and above</td><td>2 credits</td></tr></table>(Up to 2 credits)</td></tr>
<tr><th>ENRB 4-4 THERMAL COMFORT</th><th>GREENRE CREDITS</th></tr>
<tr><td>a. Ensure the consistent indoor conditions for thermal comfort:
Indoor dry-bulb temperature between 23°C to 26°C
Relative humidity between 50% to 70%</td><td>1 credit</td></tr>
<tr><td>b. Controllability of temperature.</td><td>1 credit</td></tr>
</table>

ENRB 4-5 INTERNAL NOISE LEVEL	GREENRE CREDITS
Building is designed to achieve ambient internal noise level as specified: • 55 dB (6am – 10pm) LAeq • 45 dB (10pm – 6am) LAeq	1 credit
PART 4 – INDOOR ENVIRONMENTAL QUALITY CATEGORY SCORE:	Sum of GreenRE credits obtained from ENRB 4-1 to 4-5

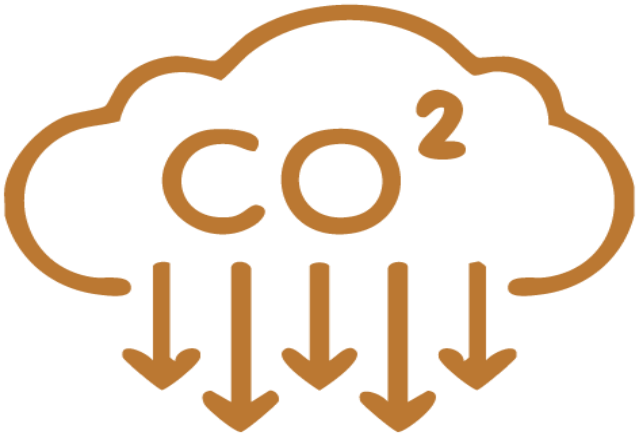


Part 5

Other Green Features

Part 5 – Other Green Features

ENRB 5-1 GREEN FEATURES & INNOVATIONS	GREENRE CREDITS
Encourage the use of green features that are innovative or/and have positive environment impact. Examples: • Vertical greening • Green Lease • Ultraviolet light-C band (UV) emitters in air handling units (AHUs) to improve indoor air quality • Provision of car park guidance system • Use of self-cleaning façade system • Use of grey water recycling system • Titanium Dioxide coating to remove odour in toilets • Use of pneumatic waste collection system Use of double refuse chutes for separating recyclable from non-recyclable waste	2 credits for high impact item 1 credit for medium impact item 0.5 credit for low impact item (Up to 10 credits)
PART 5 – OTHER GREEN FEATURES CATEGORY SCORE:	Sum of GreenRE credits obtained from ENRB 5-1



Part 6

Carbon Footprint of Development

Part 6 – Carbon Emission of Development

NRB 6-1 CARBON EMISSION OF DEVELOPMENT	GREENRE CREDITS
a. Recognise the carbon emission based on operational carbon footprint computation of the building comprising energy [B6] and water consumption [B7].	1 credit
b. Calculation of product stage embodied carbon based on following building materials [A1-A3]: • Concrete • Steel • Bricks • Cement • Others : Aluminium / Glass / Timber	1 credit
PART 6 – CARBON EMISSION OF DEVELOPMENT CATEGORY SCORE:	Sum of GreenRE credits obtained from ENRB 6-1

GreenRE Score (Existing Non-Residential Building)

GreenRE Score (ENRB) = ∑Category score [(Part 1-Energy Efficiency) +
(Part 2-Water Efficiency) +
(Part 3-Sustainable Operation & Management) +
(Part 4-Indoor Environmental Quality) +
(Part 5-Other Green Features) +
(Part 6-Carbon Emission Development)]

Where:

Category Score for Part 1≥ 30 credits and
Σ Category score for Part 2, 3, 4, 5 & 6 ≥ 20 credits

ENERGY EFFICIENCY COMPLIANCE

OPTION 1 – MINIMUM DESIGN SYSTEM EFFICIENCY (FIXED METRIC)

REQUIREMENTS

Minimum Design System Efficiency/Operating System Efficiency (DSE/OSE)

- For buildings using Water-Cooled Chilled Water Plant

GreenRE Rating	Building Cooling Load (RT)	
	< 500	≥ 500
	Efficiency (kW/RT)	
Bronze	0.85	0.75
Silver	0.80	0.70
Gold	0.75	0.68
Platinum	0.70	0.65

- For buildings using Air-Cooled Chilled Water Plant or Unitary Air-Conditioner

GreenRE Rating	Building Cooling Load (RT)	
	< 500	≥ 500
	Efficiency (kW/RT)	
Bronze	1.1	1.0
Silver	1.0	1.0
Gold	0.85	Case by case (i)
Platinum	0.78	

For building with building cooling load of more than 500RT, the use of air cooled central chilled water plant or other unitary air-conditioners are not encouraged for Gold and Platinum ratings. In general, the system efficiency of the air cooled central chilled-water plant and other unitary air-conditioners are to be comparable with the stipulated efficiency for water-cooled central chilled-water plant. Buildings that are designed with air cooled systems and for higher GreenRE rating will be assessed on a case-by-case basis.

For Gold & Platinum, the project also needs to provide the permanent measuring instruments for monitoring of water-cooled chilled-water system and air-cooled chilled water system operating system efficiency. The installed instrumentation shall have the capability to calculate resultant plant operating system efficiency (i.e., kW/RT) within 5% of its true value and in accordance with ASHRAE Guide 22 and AHRI 550/590. Heat balance test for water-cooled chilled water system is required for verification of the accuracy of the Measurement and Verification (M&V) instrumentation

DOCUMENTARY EVIDENCES

Actual Assessment:

- Detailed report from simulation software.

Site Verification Assessment

- Scenario 1), based on utility bill, if the occupancy rate is low, e.g. only 20% occupancy rate, it needs to be projected to 80% to get the BEI which reflects the actual operation situation;
- Scenario 2), based on the utility bills, If the actual operation hours are the same as what were used during the design stage, no adjustment required for operational hours; If fixed operational hours were used during design and they are different from actual operation hours, adjustment needs to be done based on actual operational hours

OPTION 2 – BUILDING ENERGY INTENSITY (BEI)

REQUIREMENTS

Total Building annual energy consumption over the gross floor area of the building (kWh/m^2/yr). Based on:

- Energy Calculation and measured data (Retrofit)
- Measurement – In operation

The project shall demonstrate the Building Energy Intensity (BEI) and show compliance to the table below (minimum occupancy >80%):

Building Type	Bronze (kWh/m^2/year)	Silver (kWh/m^2/year)	Gold (kWh/m^2/year)	Platinum (kWh/m^2/year)
Office Buildings (Small) (GFA < 15,000sqm)	205	180	135	120
Office Buildings (Large) (GFA > 15,000sqm)	210	190	155	140
Retail Mall	360	315	230	210
School, University and College	170	145	110	200
School, University and College (MOE)	60	50	40	35
Community Centres	210	190	150	125
Civic Buildings	120	105	80	70
Cultural Institutions	240	210	180	140
Sport and Recreations Centres	135	120	110	80

Table 2: Building Energy Intensity (BEI) Benchmarking

Building Type	Bronze (kWh/m^2/year)	Silver (kWh/m^2/year)	Gold (kWh/m^2/year)	Platinum (kWh/m^2/year)
Office Buildings (Small) (GFA < 15,000sqm)	150	125	100	90
Office Buildings (Large) (GFA > 15,000sqm)	130	120	90	80
Retail Mall	230	200	160	140
Community Centres	150	130	100	90
Civic Buildings	70	60	50	45
Cultural Institutions	160	135	115	100
Sport and Recreations Centres	100	85	70	65

Table 2: Building Energy Intensity (BEI) Benchmarking - District Cooling System (DCS)

DOCUMENTARY EVIDENCES

Actual Assessment:

- Details report from simulation software

Site Verification Assessment

- Scenario 1), based on utility bill, if the occupancy rate is low, e.g. only 20% occupancy rate, it needs to be projected to 80% to get the BEI which reflects the actual operation situation;
- Scenario 2), based on the utility bills, If the actual operation hours are the same as what were used during the design stage, no adjustment required for operational hours; If fixed operational hours were used during design and they are different from actual operation hours, adjustment needs to be done based on actual operational hour

WORKED EXAMPLE

Example 1:

Office building	Result based on projected data for existing building undergoing retrofit	
TBEC (Total Annual Building Energy Consumption):	250,500	kWh/year
GFA	2,584	m^2
BEI	96.9	$kWh/m^2/year$

Based on the requirement, it meets the Platinum BEI <120 requirement.

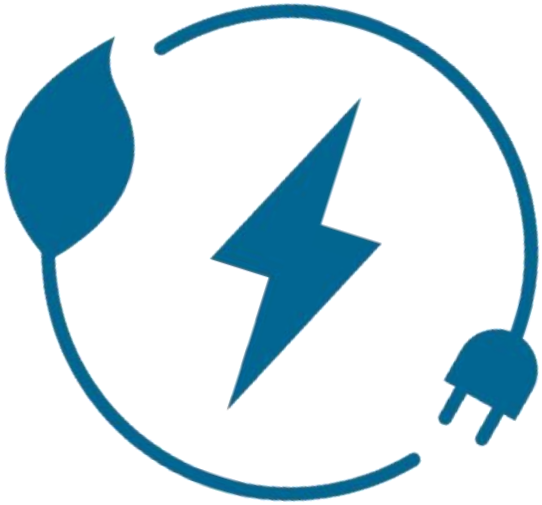
Example 2:

During verification stage, which is one year after operation, if the occupancy rate is only 30%:

A small office building	Result based on projected data for existing building undergoing retrofit	
Annual Total Building Energy Consumption: At 30% occupancy	100,000	kWh/year
GFA	2,584	m^2
BEI	38.77	$kWh/m^2/year$
TBEC Adjusted to 80% occupancy	=100,000÷ 0.3 x 0.8 = 266667	kWh/year
BEI after adjustment	103.2	$kWh/m^2/year$

With adjusted BEI of 103.2 the project can pass the verification requirement for Platinum

Energy Related Requirements



Part 1 – Energy Efficiency

ENRB 1-1 Thermal Performance of Building Envelope-OTTV

ENRB 1-2 Air-Conditioning System

ENRB 1-3 Natural Ventilation / Mechanical Ventilation

ENRB 1-4 Artificial Lighting

ENRB 1-5 Ventilation in Carparks

ENRB 1-6 Ventilation in Common Areas

ENRB 1-7 Lifts and Escalators

ENRB 1-8 Energy Efficient Practices & Feature

ENRB 1-9 Energy Policy & Management

ENRB 1-10 Renewable Energy

ENRB 1-1 THERMAL PERFORMANCE OF BUILDING ENVELOPE - OTTV

OBJECTIVES

Enhance overall thermal performance of building envelope to minimise heat gain thus reducing the overall cooling load requirement.

APPLICABILITY

Applicable to air-conditioned building spaces with aggregate areas ≥ 1000m^2.

BASELINE STANDARD

Maximum permissible OTTV = 50 W/m^2

OTTV stands for Overall Thermal Transfer Value.

Maximum permissible RTTV = 25 W/m^2

RTTV stands for Roof Thermal Transfer Value.

In the case of an air-conditioned building, the concept of Roof Thermal Transfer Value (RTTV) is applied if the roof is provided with skylight and the entire enclosure below is fully air-conditioned.

The computation of OTTV & RTTV shall be based on the methodology specified in the MS 1525:2019.

REQUIREMENTS

Up to 5 credits can be scored for building envelope with better thermal performance than the baseline standard:

0.5 credits for every reduction of 1 W/m^2 in OTTV from the baseline.

Credits scored = 0.5 x [50 - OTTV] where OTTV ≤ 50 W/m^2

For developments consisting of more than one building, the weighted average of the OTTVs based on the façade areas of these buildings shall be used as the basis for credits allocation.

That is:

$OTTV_{weighted\ average} = \sum (OTTV_{bldg} \times A_{bldg}) / A_{devt}$

where $OTTV_{bldg}$ = OTTV for building (W/m^2)
A_{bldg} = Summation of all façade areas (m^2) in a building
A_{devt} = Summation of total applicable façade areas of all buildings within the development (m^2) (i.e. $\sum A_{bldg}$)

DOCUMENTARY EVIDENCES

- Site plan with clearly demarcated the orientation of the building.
- Architectural elevation drawings showing the composition of the different façade or wall systems that are relevant for the computation of OTTV.
- Glazing specification showing the U Value and SC Value.
- Window and door schedule.
- Detailed area (m2) tabulation of fenestration and wall for every façade.
- Calculation of U Value for all type of external walls.
- Calculation of the Shading Coefficient for external shading device.
- OTTV calculation for each facing wall.
- A drawing showing the cross-sections of typical parts of the roof construction, giving details of the type and thickness of basic construction materials, insulation and air space.
- The U-value of the roof assembly and technical specification of the roof insulation (if any).

In the case of an air-conditioned building, the concept of Roof Thermal Transfer Value (RTTV) is applied if the roof is provided with skylight and the entire enclosure below is fully air-conditioned.

- RTTV Calculation (if applicable)
- Skylight specification showing the U Value and SC Value.

REFERENCES

MS1525:2019- Energy Efficiency and use of renewable energy for non-residential building - Code of Practice

ENRB 1-2 AIR-CONDITIONING SYSTEM

OBJECTIVES

Encourage the use of better efficient air-conditioned equipment to minimise energy consumption.

APPLICABILITY

Applicable to air-conditioned building areas where its aggregate air-conditioned areas ≥ 1000m^2.

Scope covers all air-conditioned equipment for the buildings as listed:

- Chillers
- Chilled water pumps
- Condenser water pumps
- Cooling Towers
- Air Handling Units (AHU)
- Fan Coil Units (FCU)
- Unitary Air-Conditioners/ Condensing Units which include single-split units, multi-spilt units and variable refrigerant flow (VRF) system

BASELINE STANDARD

Minimum efficiency requirement of the air-conditioning system stated in MS 1525:2019 or SS 530 & SS CP 13.

Water-Cooled Chilled Water Plant

Baseline	Building Cooling Load	
	< 500 RT	≥ 500 RT
Prerequisite Requirements Minimum system efficiency of central chilled-water plant	0.85 kW/RT	0.75 kW/RT

i. Water-Cooled Chiller – Refer Table 25 of MS 1525:2019 to calculate Its Coefficient of Performance (COP)

ii & iii. Chilled-water pump and condenser water pump efficiency – Refer to Clause 8.2.5 in MS 1525:2019 which states that for chilled water or condenser water pumping system operating for more than 750 hours a year, the pump efficiency shall be:

Type of pumping system	Maximum Power consumption [W/(m3/h)]
Condenser water pump	84
Chilled water pump	97

Table 21. Maximum power consumption for pumping system

- Cooling tower performance at the rating condition states in Table 3 SS 530.

Rating condition is as follows: 35°C Entering water
29°C Leaving water
24°C Wet Bulb Outdoor air

Propeller and axial fan cooling tower:

With heat rejected from every 3.23 L/s of condenser water per 1 kW of fan power rating:

Cooling tower performance ≤ 1kW / 3.23 L/s
≤ 0.310 kW/ L/s

Centrifugal fan cooling tower:

With heat rejected from every 1.7L/s of condenser water per 1kW of fan power rating:

Cooling tower performance ≤ 1kW / 1.7 L/s
≤ 0.588 kW / L/s

OR

Air-Cooled Chilled-Water Plant / Unitary Air-Conditioners

Baseline	Building Cooling Load	
	< 500 RT	≥ 500 RT
Prerequisite Requirements Minimum system efficiency of air cooled chilled water plant or unitary conditioners	1.1 kW/RT	1.0 kW/RT

- Air-cooled chilled water plant - Refer Table 25 of MS 1525:2019 to calculate its Coefficient of Performance (COP).
- Unitary Air-Conditioners / Condensing Units – Refer Table 23 of MS 1525:2014

Note: If the specific type of air conditioned is not found in MS 1525:2019, please refer to SS 530 to make the calculation on COP. Priority given to MS 1525:2019.

Air Distribution System – Refer ASHRAE 90.1:2010 Clause 6.5.3.1 as prescribed below:

Baseline Air Distribution System Type	Allowable Fan System Input Power	
	($kW/m^3/s$)	(W/CMH)
AHUs / FCUs ≥ 4kW (Constant Volume)	1.5	0.42
AHUs ≥ 4kW (Variable Volume)	2.1	0.58
Fan systems with nameplate motor power < 4kW	0.6	0.17

Provision of permanent measuring instruments to monitor water-cooled and air-cooled chilled water plant

- The instrumentation installed in the system shall have capability to calculate resultant plant efficiency within ± 5% of its true value – Refer ASHRAE Guide 22 and AHRI 550/590.
- To ensure the monitoring of central water-cooled chilled plant efficiency remains within a ± 5% margin of error, the following instrumentation accuracy is recommended;

Description	Measurement error
Temperature sensors • 10K/30K Thermistor • Platinum Resistance Thermometers	± 0.03 – 0.05 °C at 0°C
Flow Sensor Meter • Ultrasonic • Full bore magnetic	± 0.5 – 1.0 % over entire measurement range
Power meter	ANSI C12.1-2008, Class 1 ±1%

Verification of central chilled water plant instrumentation – Heat Balance substantiating test

- Substantiating test shall be conducted as accordance to AHRI 550/590
- The heat balance shall be conducted over entire normal operating hours with more than 80% of the computed balance within ± 5% over the audit period

Heat balance is denoted by below equation:

$q_{condenser} = q_{evaporator} + W_{input}$

Where;

$q_{condenser}$ = heat rejected (in kW or RT)
$q_{evaporator}$ = cooling load (in kW or RT)
W_{input} = measured electrical power input to compressor

1-2(f)

Provisioning of variable speed controls for chiller plant equipment

1-2(g)

Provisioning of automatic control devices or sensors to regulate outdoor air flow rate to maintain the concentration of Carbon Dioxide at acceptable range ≤700 ppm above outdoor concentration.

REQUIREMENTS

1-2 Option 1: Fixed Metric

a. Water-Cooled Chilled-Water Plant (Up to 20 credits)

- **Building cooling load ≥ 500RT:**

  14 credits for achieving plant efficiency of 0.75 kW/RT

  0.35 credit for every percentage improvement in the chiller plant efficiency better than 0.75 kW/RT

  Credit scored = 0.35 x (% improvement)

- **Building cooling load < 500RT:**

  14 credits for achieving plant efficiency of 0.85 kW/RT

  0.3 credit for every percentage improvement in the chiller plant efficiency better than 0.85 kW/rt

  Credit scored = 0.3 x (% improvement)

  (up to 20 credits)

OR

b. Air Cooled Chilled-Water Plant / Unitary Air-Conditioners (Up to 20 credits)

- **Building cooling load ≥ 500RT:**

  14 credits for achieving plant efficiency of 1.0 kW/RT

  0.25 credit for every percentage improvement in the chiller plant efficiency better than 1.0 kW/RT

  Credit scored = 0.25 x (% improvement)

- **Building cooling load < 500RT:**

  14 credits for achieving plant efficiency of 1.1 kW/RT

  0.2 credit for every percentage improvement in the chiller plant efficiency better than 1.1 kW/RT

  Credit scored = 0.2 x (% improvement)

  (up to 20 credits)

c. Air Distribution System (Up to 8 credits)

0.15 credits for every percentage improvement in the air distribution system efficiency above the baseline.

Credits scored = 0.15 x (% improvement)

d. 2 credit can be scored for the provision of permanent measuring instruments for monitoring of water cooled chilled-water plant and air-cooled chilled water plant efficiency

e. 1 credit can be scored for verification of central water cooled chilled-water plant instrumentation: Heat Balance – substantiating test for water cooled chilled-water plant to be computed in accordance with AHRI 550/590. The operating system efficiency and heat balance to be submitted to GreenRE upon commissioning.

f. 1 credit can be scored if variable speed controls for chiller plant equipment such as chilled-water pumps and cooling tower fans are provided to ensure better part-load plant efficiency.

g. 1 credit can be scored if sensors or similar automatic control devices are used to regulate outdoor air flow rate to maintain the concentration of carbon dioxide (CO_2) ≤ 700 ppm above outdoor.

1-2 Option 2: BEI Benchmarking

i. Projects with air conditioning system

a. 14 credits for achieving BEI per table shown.

Building Type	Bronze ($kWh/m^2/year$)	Silver ($kWh/m^2/year$)	Gold ($kWh/m^2/year$)	Platinum ($kWh/m^2/year$)
Office Buildings (Small) (GFA < 15,000sqm)	205	180	135	120
Office Buildings (Large) (GFA > 15,000sqm)	210	190	155	140
Retail Mall	360	315	230	210
School, University and College	170	145	110	200
School, University and College (MOE)	60	50	40	35
Community Centres	210	190	150	125
Civic Buildings	120	105	80	70
Cultural Institutions	240	210	180	140
Sport and Recreations Centres	135	120	110	80

Table 3: Building Energy Intensity (BEI) Benchmarking (kwh/m2/yr)

b. Water-Cooled Chilled-Water Plant (Up to 8 credits)

- **Building cooling load ≥ 500RT:**

  0.35 credit for every percentage improvement in the chiller plant efficiency better than 0.75 kW/RT

  Credit scored = 0.35 x (% improvement)

- **Building cooling load < 500RT:**

  0.3 credit for every percentage improvement in the chiller plant efficiency better than 0.85 kW/RT

  Credit scored = 0.3 x (% improvement)

c. Air Cooled Chilled-Water Plant / Unitary Air-Conditioners (Up to 8 credits)

- **Building cooling load ≥ 500RT:**

  0.25 credit for every percentage improvement in the chiller plant efficiency better than 1.0 kW/RT

  Credit scored = 0.25 x (% improvement)

- **Building cooling load < 500RT:**

  0.2 credit for every percentage improvement in the chiller plant efficiency better than 1.1 kW/RT

  Credit scored = 0.2 x (% improvement)

  (up to 8 credits)

d. 2 credits can be scored for the provision of permanent measuring instruments for monitoring of water cooled chilled-water plant and air-cooled chilled water plant efficiency

e. 1 credit can be scored for verification of central water cooled chilled-water plant instrumentation: Heat Balance – substantiating test for water cooled chilled-water plant to be computed in accordance with AHRI 550/590. The operating system efficiency and heat balance to be submitted to GreenRE upon commissioning.

f. 1 credit can be scored if variable speed controls for chiller plant equipment such as chilled-water pumps and cooling tower fans are provided to ensure better part-load plant efficiency.

g. 1 credit can be scored if sensors or similar automatic control devices are used to regulate outdoor air flow rate to maintain the concentration of carbon dioxide (CO_2) ≤ 700 ppm above outdoor.

ii. Project with District Cooling System

a. 20 credits for achieving BEI per table shown

Building Type	Bronze (kWh/m^2/year)	Silver (kWh/m^2/year)	Gold (kWh/m^2/year)	Platinum (kWh/m^2/year)
Office Buildings (Small) (GFA < 15,000sqm)	150	125	100	90
Office Buildings (Large) (GFA > 15,000sqm)	130	120	90	80
Retail Mall	230	200	160	140
Community Centres	150	130	100	90
Civic Buildings	70	60	50	45
Cultural Institutions	160	135	115	100
Sport and Recreations Centres	100	85	70	65

Table 2: Building Energy Intensity (BEI) Benchmarking - District Cooling System (DCS) (kwh/m2/yr)

b. Air Distribution System (Up to 8 credits)

0.15 credits for every percentage improvement in the air distribution system efficiency above the baseline.

Credits scored = 0.15 x (% improvement)

DOCUMENTARY EVIDENCES

For 1-2 Option 1 (a) & (b) and Option 2 (b) & (c)

- Detailed calculations of the overall improvement in equipment/system efficiency of the air-conditioning plants/ showing the design cooling system capacity and the system efficiency (including individual equipment efficiency).
- Calculation and technical data of the designed system efficiency of chillers at part load condition.
- Technical product information of all air-conditioning and system which included chillers, chilled water pumps, condenser water pumps, cooling towers.
- Schematic drawings showing the air-conditioning system
- Schedules of the air-conditioning system.
- Operating System Efficiency (OSE) report during the Site Verification Assessment (SVA)

For 1-2 Option 1 (c) and Option 2 (ii) District Cooling (b)

- Detailed calculations of the overall improvement for air distribution system.
- Technical product information of all AHUs, FCUs, and etc.
- AHUs and FCUs schedule and schematic drawing.

For 1-2 Option 1 and Option 2 (d)

- Instrument's calibration certificates from accredited laboratory or batch calibration certificates from manufacturer.
- Schematic drawing showing the location of the digital power meters, flow meters and temperature sensors.
- Summary of instruments, standard and measurement accuracy to be presented in the following format:
- Technical specification of the digital power meters, flow meters and temperature sensors.

For 1-2 Option 1 and Option 2 (e)

- Computation of the percent heat balance that is the total heat gain and total heat rejected must be within ± 5% for 80% of the sampled credits over the normal building operations hours accordance with AHRI550/590.
- Detailed calculations of the overall uncertainly of measurement of the resultant chiller plant efficiency in kW/RT to be within ± 5% of the true value based on instrumentation specification. Uncertainly
- Measurement and Verification report signed by competent person.

For 1-2 Option 1 and Option 2 (f) and (g)

- Extracts of the tender specification showing the requirements to incorporate these control devices.
- Plan layouts showing the locations and the types of control devices used to regulate fresh air intake.
- Technical product specification of the control devices.

For Option 2 (i) Air Conditioning System (a) and (ii) District Cooling system (a)

- Tabulation data for the energy consumption of the building for the 1 year.
- Building Energy Intensity (BEI) calculation

REFERENCES

a. MS 1525:2019 – Energy efficient and use of renewable energy for non-residential building – Code of Practice

b. SS 530 – Code of Practice for Energy Efficiency Standard for Building Services and Equipment.

c. SS CP 13 – Code of Practice for Mechanical Ventilation and Air-Conditioning in Buildings.

ENRB 1-3 NATURAL VENTILATION / MECHANICAL VENTILATION

OBJECTIVES

Encourage building that facilitates good natural ventilation. Encourage energy efficient mechanical ventilation system as the preferred ventilation mode to air-conditioning in buildings.

APPLICABILITY

Applicable to Non-Air-Conditioned Building Areas (with an aggregate non-air-conditioned areas > 10% of total floor area excluding carparks and common areas) for Natural Ventilation.

BASELINE STANDARD

Fan power limitation in mechanical ventilation systems:

Allowable nameplate motor power	
Constant volume	Variable Volume
1.7 kW/m^3/s	2.4 kW/m^3/s

REQUIREMENTS

1-3(a) Natural Ventilation

Up to 32 credits will be awarded for natural ventilation in the building.

20 base credits will be awarded for use of natural ventilation,

Up to 12 credits can be scored for building design that utilises prevailing wind conditions to achieve adequate cross ventilation.

1.2 credit for every (10% of units/ rooms with window openings facing north and south directions)

Credits scored = 1.2 x (% of units / 10)

Note: In Malaysia, the prevailing wind comes from two predominant directions; that is the north to north-east during the Northeast monsoon season and south to south-east during the South-west monsoon season. Hence, buildings designed with window openings facing the north and south directions have the advantages of the prevailing wind conditions which would enhance indoor thermal comfort. Meteorological data on the more precise wind direction and velocity of the site location can also be used as the basis for the design.

It is not necessary for the window openings to be located perpendicularly to the prevailing wind direction. Only window adjoining the habitable space to be considered. Window adjoining toilets/ bathroom and store room will not be considered.

An oblique angle is considered acceptable (see illustrations as shown in the next page).

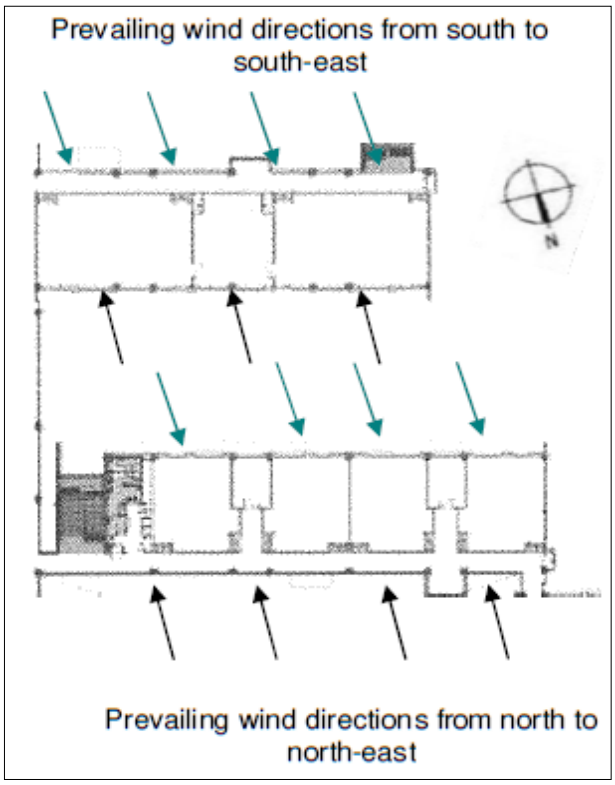


Illustration 1-3.1: Orientation of facade facing North and South

Building layout shows all rooms with window openings facing the North and South directions.

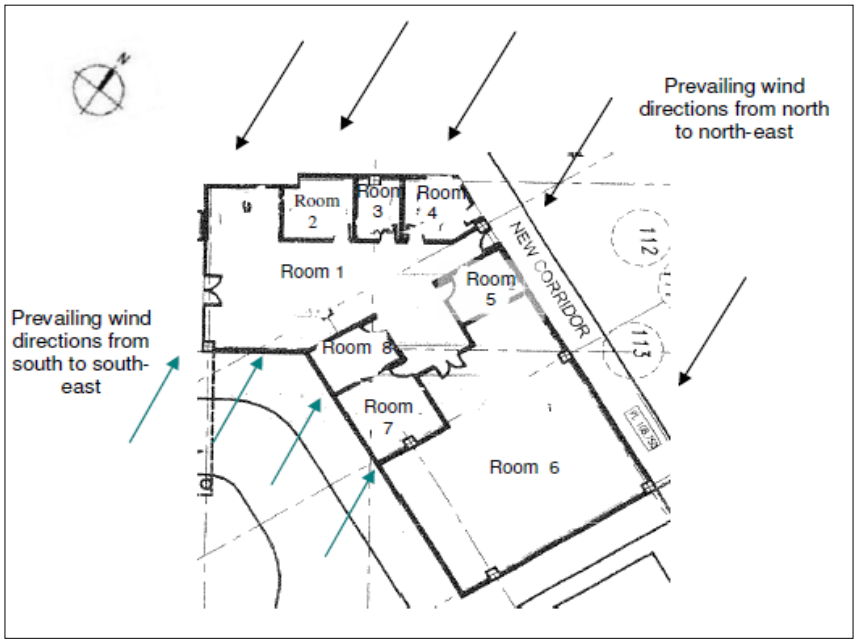


Illustration 1-3.2: Orientation of facade facing North and South

Building layout shows not all rooms with window facing the north and south directions. Room 2 to Room 5 would only have prevailing wind in one direction. Only Room 1 and 6 can be considered meeting requirement 1-3(a).

1-3(b) Mechanical Ventilation

Up to 32 credits for the use of mechanical system in order to promote adequate ventilation between indoor and outdoor air.

0.6 credits for every subsequent 1% improvement in the baseline.

DOCUMENTARY EVIDENCES

1-3(a) Natural Ventilation

- Architectural plan layouts showing the units / rooms of all blocks with highlights of those with north and south window openings.
- Calculation showing the percentage of units or rooms with window openings facing north and south directions in the prescribed formats as shown in Table 1-3(a).

Ref	Description	Units/Rooms with window opening in the N-S direction (a)	Total no. of naturally ventilated units/room (b)	% of units/ rooms with window opening in N-S direction
1	Classroom Blk A & A1			∑ (a) / ∑(b) x 100
2	Classroom Blk B			
3	Offices, meeting rooms and computer rooms with air-conditioning			
	Total:			

Table 1-3(a) - Percentage of units with window opening in N-S direction

Credits scored = 1.2 x (% of units / 10)
= 1.2 x [(∑ (a) / ∑(b) x 100) / 10] + 20 (for use of NV)

1-3(b) Mechanical Ventilation

- Plan layout demarcates the area with mechanical ventilation system.
- The overall design and drawings for mechanical ventilation system to make up the required outdoor air quantity into the building at desire fan power limit.
- Detailed calculations showing the fan power improvement.
- Product catalogue of the fan power used.

REFERENCES

SS CP 13 – Code of Practice for Mechanical Ventilation and Air-Conditioning in Buildings

ENRB 1-4 ARTIFICIAL LIGHTING

OBJECTIVES

Encourage the use of better efficient lighting to minimise energy consumption from lighting usage while maintaining proper lighting level.

APPLICABILITY

Applicable to lighting provisions that designed in accordance to the luminance level as recommended in MS 1525:2019.

BASELINE STANDARD

Luminance level stated in MS 1525:2019 –Energy Efficiency and Use of Renewable Energy for Non-Residential Building – Code of Practice.

REQUIREMENTS

Up to 13 credits if tenants' light is provided **OR** up to 5 credits if tenants' light is excluded for the improvement in the lighting power consumption.

0.25 credit for every percentage improvement in the lighting provisions over the baseline standard. That is:

Credits scored = 0.25 x (% improvement)

Display lighting and specialised lighting are to be included in the calculation of lighting power budget.

The design service illuminance, lamp efficiencies and the light output ratios of luminaries shall be in accordance with in MS 1525:2019 – Energy Efficiency and Use of Renewable Energy for Non-Residential Building - Code of Practice.

DOCUMENTARY EVIDENCES

- Lighting layout plan.
- Lighting schedules showing the numbers, locations and types of luminaries used.
- Calculation of the installed lighting power budget and the percentage, improvement in the prescribed tabulated format.
- Technical product information of the lighting luminaries used.

REFERENCES

MS 1525:2019 –Energy Efficiency and Use of Renewable Energy for Non-Residential Building – Code of Practice

ENRB 1-5 VENTILATION IN CARPARKS

OBJECTIVES

Encourage the use of energy efficient design and control of ventilation systems on carparks.

APPLICABILITY

Applicable to all carpark's spaces in the development.

REQUIREMENTS

1-5(a)

4 credits can be scored for car park spaces that are fully naturally ventilated.

1-5(b)

For carparks that have to be mechanically ventilated, credits can be scored for the use of carbon monoxide (CO) sensors in regulating such demand based on the mode of mechanical ventilation (MV) used; 2.5 credits for carparks using fume extract system and 2 credits for those with MV with or without supply.

Note: When there is a combination of different ventilation modes adopted for carpark design, the credits scored under this requirement will be prorated accordingly.

DOCUMENTARY EVIDENCES

For 1-5(a)

- Plan layouts showing all car park provision for the development with highlights of the car park spaces that are designed to be naturally ventilated.
- Calculation showing the openings at the carpark level to meet the UBBL requirement.

For 1-5(b)

- Plan layouts showing all car park provision for the development with highlights of the car park spaces that are designed to be naturally ventilated and/or mechanical ventilated.
- Plan layout indicating the location of CO sensors and the mode of ventilation adopted for the design.
- Calculation showing the credits allocation if there is a combination of different ventilation mode adopted for the car park design.
- Technical product information of CO sensors and mechanical ventilation.

ENRB 1-6 VENTILATION IN COMMON AREAS

OBJECTIVES

Encourage the use of energy efficient design and control of ventilation systems in common areas.

APPLICABILITY

Applicable to the following common areas of the development.

- Toilets
- Staircases (include BOMBA staircase)
- Corridors
- Lift Lobbies
- Atriums

REQUIREMENTS

Up to 5 credits can be scored for the use of natural ventilation as an effective passive cooling design strategy to reduce the energy used by air- conditioning systems in these common areas.

Credits are scored based on the mode of ventilation provided in these applicable areas.

Natural ventilation – 1.5 credits for each area

Mechanical ventilation – 0.5 credit for each area

DOCUMENTARY EVIDENCES

- Plan layouts showing the applicable areas and the respective modes of ventilation with proper demarcation of the opening.
- Schedules showing the numbers, locations of the applicable areas and the mode of ventilation used.
- Technical product information of mechanical ventilation system. (if applicable)
- Schematic drawing of the mechanical ventilation system.
- Calculation showing the credits allocation if there is a combination of different ventilation modes adopted for the applicable areas.

ENRB 1-7 LIFTS AND ESCALATORS

OBJECTIVES

Encourage the use of energy efficient lifts and escalators.

APPLICABILITY

Applicable to **ALL** lifts and/or escalators in the development.

REQUIREMENTS

1 credit can be scored for the use of lifts with energy efficient features such as sleep mode or regenerative features or motorless lift.

1 credit can be scored for the use of escalators with motion sensors to regulate usage.

DOCUMENTARY EVIDENCES

- Extracts of the tender specification indicating the types of lifts & escalators and related features used.
- Plan layout showing the location of the lifts and escalators.
- Schedules showing the total number of lifts & escalators and its power consumption.
- Technical information of the lifts & escalators.

WORKED EXAMPLE 1-9

Proposed development has the following provision:

Two lift types : Type L1 with sleep mode features, Type L2 with sleep mode features
Two escalator types : Type E1 with motion sensors, Type E2 without motion sensors

1 credit for the use of lifts with sleep mode features
No credits for escalators as **not all** escalators are designed with motion sensors
Credits scored for 1-9 = 1 credits (out of 2 credits)

ENRB 1-8 ENERGY EFFICIENT PRACTICES & FEATURES

OBJECTIVES

Encourage the use of energy efficient practices and features which are innovative and/or have positive environmental impact.

APPLICABILITY

Applicable to practices and features that are not listed in the requirements under Part 1 – Energy Efficiency.

REQUIREMENTS

1-8(a)

1 credit can be scored for the practice of using Building Energy Intensity (BEI) as a building performance indicator to measure the building's unit area energy consumption for future monitoring and improvements.

BEI is derived using the following equation:

BEI = [(TBEC – CPEC) / (GFA excluding carpark - GLA x FVR) x (NF/OH)

Where:

TBEC = Total building energy consumption excludes renewable energy replacement (kWh/year)
CPEC = Car Park Energy Consumption in (kWh/year)
GFA = Gross Floor Area (exclude car park area) (m^2)
GLA = Gross Lettable Area (m^2)
FVR = Floor Vacancy Rate (NLA) (m^2)
NF = Normalizing factor based on a typical weekly operating hour that is 52 Hrs/week [only for office category]
OH = Weighted weekly operating hours (hrs/week) [only for office category]

1-8(b)

0.5 credits can be scored for each equipment type used up to 2 credits.

1-8(c)

Up to 9 credits can be scored for this section. 2 credits for every 1% energy saving over the total building energy consumption.

Examples include:

- Re-generative lift
- Heat recovery system
- Motion sensors
- Sun pipes
- Light shelves
- Photocell sensors to maximise the use of Daylight
- Heat pumps, etc.

Note: For features that are not listed NRB 1-8(b) above, the QP is required to submit the details showing the positive environmental impacts and potential energy savings of the proposed features to GreenRE assessment.

DOCUMENTARY EVIDENCES

For 1-8(a)

- Calculation of the Building Energy Intensity (BEI) using the pre-determined daily usage pattern.
- Detail calculation including operation hours for the estimated energy load for each component in the building etc.: lighting, air conditioning system, pump, receptacle load.
- Technical product information and related drawing on the energy efficient features.
- List of the assumption for the BEI calculation.

For 1-8(b)

- Extracts of the tender specification showing the provision of the proposed energy efficient products and the extent of implementation where applicable.
- Technical product information and certificate.

For 1-8(c)

- Extracts of the tender specification showing the provision of the proposed energy efficient features and the extent of implementation where applicable.
- Technical product information and related drawing on the energy efficient features used.
- Calculation of the percentage energy saving that could be reaped from the use of these features.

ENRB 1-9 ENERGY POLICY & MANAGEMENT

OBJECTIVES

Encourage new strategies and plans in the future save and minimise the use of energy.

APPLICABILITY

Applicable to building that uses energy.

REQUIREMENTS

1-9(a)

0.5 credits for energy policy, energy targets and regular review with top management's commitment as part of an environmental strategy.

1-9(b)

0.5 credits to show intent, measures and implementation strategies of energy efficiency improvement plans to achieve energy target set over the next three years. Committed energy savings accrued from proposed measures should be quantified

DOCUMENTARY EVIDENCES

For 1-9(a)

- Energy policy showing energy saving commitments or energy targets from the top management.

For 1-9(b)

- Improvement plans showing the calculation of energy saving that can be achieved over the next three years.

ENRB 1-10 RENEWABLE ENERGY

OBJECTIVES

Encourage the use of renewable energy sources in buildings.

APPLICABILITY

Includes all renewable energy sources

REQUIREMENTS

Up to 20 credits can be scored for the use of renewable energy. Credit scored based on % replacement of electricity by renewable energy source

5 credits for every 1% replacement of electricity (based on total electricity consumption) by renewable energy.

OR

3 credits for every 1% replacement of electricity (based on total electricity consumption excluding tenant's usage) by renewable energy.

OR

3 credits for every 10% of roof area used for solar panels.
1 credit for solar ready roof

Note:

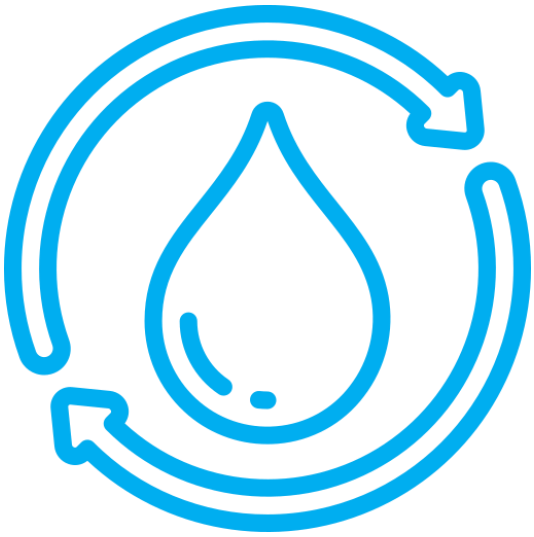
- Projects that implement Green Energy Tariff (GET) purchases are not eligible to score under this credit. GET is only applicable for scoring under Part 6: Operational Carbon Footprint of Development.
- Credits can be claimed if the renewable energy system is owned and operated beyond the certification boundary, provided that the energy generated is directly attributed to the project.
- The installation of onsite renewable energy systems that are used off-site is not eligible to score under this credit. However, such installations may be considered under Part 5: Other Green Features.

Note: The credit scored for renewable energy provision shall not result in a double grade jump in GreenRE rating (i.e from GreenRE Bronze or Silver to Gold or Platinum)

DOCUMENTARY EVIDENCES

- Plan layout showing the location of proposed renewable energy system.
- Technical product information on the salient features if the renewable energy system and the expected renewable energy generated.
- Calculation of the percentage replacement of electricity and the total annual electricity consumption of the development.

Other Green Requirements



Part 2 – Water Efficiency

ENRB 2-1 Water Usage and Leak Detection

ENRB 2-2 Water Efficient Fittings

ENRB 2-3 Alternative Water Sources

ENRB 2-4 Water Efficiency Improvement Plans

ENRB 2-5 Irrigation System and Landscaping

ENRB 2-6 Water Consumption of Cooling Tower

ENRB 2-1 WATER USAGE AND LEAK DETECTION SYSTEM

OBJECTIVES

Promote the use of sub-metering and leak detection system for better control and monitoring of water usage

APPLICABILITY

Applicable to sub-metering provisions for major water uses of the building developments.

REQUIREMENTS

a. 1 credit can be scored by monitoring the water consumption on a monthly basis.

b. 1 credit can be scored for the provision of private-meters for major water uses (e.g., Cooling tower, water features, irrigation, swimming pools and tenant usage)

c. Up to 2 credits can be scored for the provision of automated/smart metering for monitoring and leakage detection.

DOCUMENTARY EVIDENCES

For 2-1 (a)

- Monitoring plan of water consumptions.

For 2-1 (b)

- Schematic drawings of cold-water distribution system showing the location of the sub meters provided.
- List of sub metering and its location.

For 2-1 (c)

- Schematic drawing showing the location of sub-metering and its linkage to the Building Management System (BMS).
- List of input and output point of the Building Management System (BMS) with highlighted the submeter point.
- Printouts of smart metering results data log for each sub-meter.

ENRB 2-2 WATER EFFICIENT FITTINGS

OBJECTIVES

Reduce the use of potable water by using water efficient fittings covered under the Water Efficiency Product Labelling Scheme (WEPLS) or Water Efficiency Labelling Scheme (WELS).

APPLICABILITY

Applicable to the water fittings covered by the WEPLS or WELS as follows:

- Basin taps and mixers
- Sink/taps and mixers
- Dual Flush Low-Capacity Flushing Cisterns
- Showerheads, Shower taps and mixers
- Urinals and Flush Valves

BASELINE STANDARD

As specified under Water Efficiency Product Labelling Scheme (WEPLS)

REQUIREMENTS

Up to 12 credits can be scored based on the number and water efficiency rating of the fitting type used.

Rating Based on Water Efficiency Products Labelling Scheme (WEPLS)		
Efficient*	Highly Efficient**	Most Efficient***
6	**9**	**12**

DOCUMENTARY EVIDENCES

- Extracts of the tender specification showing all the water fitting provisions for the development.
- Water fitting schedules showing the numbers, types and the approved rating of the proposed fittings in the prescribed tabulated format shown in the Table 2.1-1.
- Schematic drawing of cold water and sanitary plumbing.
- WEPLS product specification or certificate. In the event no product recognition from WEPLS, product catalogue and test report from local or international body that equivalent to the SIRIM standard of testing is required.

REFERENCES

For more information about WEPLS, refer to

http://www.span.gov.my/index.php?option=com_content&view=article&id=580%3Aabout-us1&catid=175%3Awepls&Itemid=457&lang=en

WORKED EXAMPLE 2-2

Example of a water fitting schedule showing the numbers, types and the approve rating of the proposed fitting for a residential development (including common facilities such as clubhouse toilets).

Ref.	Water Fitting Type	WEPLS rating			Not Rated	Total
		Efficient	Highly Efficient	Most Efficient		
1	Shower taps and mixers	0	45	0	0	45
2	Basin taps and mixers	0	0	55	0	55
3	Sink/bib taps and mixers	0	70	0	0	70
4	Flushing cisterns	0	0	50	0	50
5	Others – Unrinals for club house	0	0	0	5	5
Total no. based on rating (A)		0	115	105	5	∑A = 225
Weightage (B)		**6**	**9**	**12**	**0**	
Total (A x B)		0	1035	1260	0	∑(AxB) = 2295

Table 2-2(a) : Computation of the percentage for water fittings

Credits scored = ∑(A x B) / ∑A
= 2295 / 225
= 10.2 credits

ENRB 2-3 ALTERNATIVE WATER SOURCES

OBJECTIVES

Encourage the use of suitable systems that utilize alternative water sources for **non-potable uses**: irrigation, washing, water features, toilet flushing, etc

APPLICABILITY

Generally applicable to building that uses alternative water sources.

REQUIREMENTS

Up to 3 credits will be awarded based on the % reduction in total potable water usage of the applicable uses

> 50 %	3 credits
≥ 10 % to 50 %	2 credits
< 10 %	1 credit

DOCUMENTARY EVIDENCES

- Drawings showing the location and design of non-potable water source.
- Product information or related catalogue on the alternative water source used.
- Calculation of potable water usage of the building
- Calculation on the percentage of water reduction that can be achieved by using this alternative source.

ENRB 2-4 WATER EFFICIENCY IMPROVEMENT PLANS

OBJECTIVES

To show intent, measures and implementation strategies of water efficiency improvement plans over the next three years

APPLICABILITY

Generally applicable for water consumption and its cycle for non-domestic used

REQUIREMENTS

1 credit can be scored for the commitment to plan water savings accrued from proposed measures

DOCUMENTARY EVIDENCES

- Improvement plans showing the calculation of water saving that can be achieved.
- Water efficiency management plan report

ENRB 2-5 IRRIGATION SYSTEM AND LANDSCAPING

OBJECTIVES

Reduce potable water consumption by provision of suitable systems that utilise rainwater or recycled water for landscape irrigation.

APPLICABILITY

Applicable to development with landscaping provision.

REQUIREMENTS

2-5(a)

1 credit can be scored for the use of non-potable water including rainwater for landscape irrigation.

For rainwater harvesting tank provision, RWHT to be sized in accordance to Guideline for Rainwater Harvesting and Utilisation System (SPAH) and MSMA guidelines. The rainwater tanks are to be optimally sized to cater for outdoor water use only.

Summary calculation of % replacement of outdoor water requirements from rainwater harvesting to be provided as part of claim for this credit.

2-5(b)

1 credit can be scored if more than 50% of the landscape areas are served by water efficient irrigation system with features such as automatic sub-soil drip irrigation system with rain sensor control.

2-5(c)

1 credit can be scored if at least 50% of the landscape areas consist of drought tolerant plants or plants that require minimal irrigation.

WORKED EXAMPLE 2-5(a)

Landscape Consumption

Location	Landscape type	Water Required (L/day)	Quantity		Total watering requirement (L/Day)
GF	Tree	24	200	Nos	4800
	Shrub	6.3	5660	m2	35658
	Turf	3.1	1415	m2	4386.5
					44844.5

Irrigation water requirement (Litre/Day)	44844.5

Roof Catchment

Type	m2	Run-off coefficient
Pitched Tile		0.8
Steel Roof	1239	0.9
RC Roof	1110	0.5
Block Pavement		0.7
Gravel Roadway		0.3

Total Catchment Area (m^2) = 2349
Catchment Area x Run -off coefficient = 1670.1

Type Of System	First Flush System
Equation	Collectible Rainwater = Rainfall x Catchment Area x Run Off Coefficient - (Total Catchment Area x First Flush Diversion)
First Flush Diversion (L/sqm)	1
Tank Size (L)	160,000.00
Total Annual Collected Rain Water (L)	3,880,633.50
Average Daily Collected Rain Water (L)	10,631.87
Irrigation Consumption (L/Day)	44,844.50
Percentage of Reduction (%)	**23.71**

DOCUMENTARY EVIDENCES

For 2-5(a)

- Calculation showing the percentage of potable water saving for irrigation.
- Drawings showing the location and design of non-potable water source.
- Extracts of the tender specification showing how the non-potable water source is to be provided.

For 2-5(b)

- Drawings showing the overall landscape areas and the areas that would be served using the system.
- Calculation showing the percentage of the landscape areas that would be served using the system (at least 50%).
- Extracts of the tender specification showing the provision and details of water efficient irrigation system.
- Product technical information of the irrigation system.

For 2-5(c)

- Drawings showing the overall landscape areas and the areas that use drought tolerant plants or plants that require minimal irrigation.
- Calculation showing the percentage of the landscape areas that use drought tolerant plants or plants that require minimal irrigation (at least 80%).
- Plant species showing the minimum water requirement.

ENRB 2-6 WATER CONSUMPTION OF COOLING TOWER

OBJECTIVES

Reduce potable water consumption for cooling purpose.

APPLICABILITY

Applicable to building development with water-cooled central chillers systems and water-cooled package units.

REQUIREMENTS

2-6(a)

1 credit can be scored for the use of cooling tower water treatment system which can achieve 6 or better cycles of concentration at acceptable water quality.

2-6(b)

1 credit can be scored for the use of recycled water from approved sources to meet the water demand for cooling purpose.

DOCUMENTARY EVIDENCES

For 2-6(a)

- Extracts of the tender specification showing the requirements to incorporate with the cooling tower designs to achieve six cycles of concentration.
- Details showing how the cooling towers have been designed to achieve at least six cycles of concentration.
- Relevant drawings showing the location of the cooling towers and other supporting systems that are required to achieve the designed concentration.

For 2-6(b)

- Extracts of the tender specification showing how the recycled water source is to be provided.
- Details of the recycled water system.
- Schematic system showing the recycling system.

Other Green Requirements



Part 3 – Sustainable Operation & Management

ENRB 3-1 Building Operation & Maintenance

ENRB 3-2 Post Occupancy Evaluation

ENRB 3-3 Waste Management

ENRB 3-4 Sustainable Products

ENRB 3-5 Greenery Provision

ENRB 3-6 Environmental Protection

ENRB 3-7 Green Transport

ENRB 3-1 BUILDING OPERATION & MAINTENANCE

OBJECTIVES

Encourage the adoption of environmentally friendly practices during construction and building operation.

APPLICABILITY

Generally applicable to all building developments.

REQUIREMENTS

3-1(a)

1 credit can be scored if the environmental policy that reflects the sustainable goals set.

3-1(b)

1 credit can be scored for the provision of building users' guide with details of the environmentally friendly facilities and features within the building and their uses in achieving the intended environment performance during building operation.

The minimum requirement of the Green Building User Guide as follows;

- Details of green building certification i.e rating tier, scorecard, certificate, validity etc.
- Summary of green building features (ideally with photographs and diagrams)
- Recommended practices for enhanced environmental performance of the building
- Green fit out guidelines to details recommended minimum environmental standards to assist building users in making sustainable fit- out decisions
- GreenRE renewal guideline

3-1(c)

1 credit can be scored if the building maintenance team comprises of 1 GREAP / GMAP

3-1(d)

1 credit can be scored if the project team comprises 1 GREAP/ GMAP

3-1(e)

Up to 1 credit if the environmental management system of the building is ISO 14000 or ISO 50001 certified.

DOCUMENTARY EVIDENCES

For 3-1(a)

- Documentation related to the building environmental policy.

For 3-1(b)

- A copy of the building users' guide containing the details of the environmentally friendly facilities and features within the building and their uses in achieving the intended environment performance during building operation.

For 3-1(c)

- A certified true copy of certificate of GREAP / GMAP where applicable and a copy of organization chart of the building management team.

For 3-1(d)

- A certified true copy of certificate of GREAP / GMAP where applicable and a confirmation of their involvement in the project.

For 3-1(e)

- A certified true copy of the ISO 14000 and ISO 50001 certificate from the facilities management team.

ENRB 3-2 POST OCCUPANCY EVALUATION

OBJECTIVES

To receive feedback from occupants of the building.

APPLICABILITY

Generally applicable to all building developments.

REQUIREMENTS

3-2(a)

2 credits for the conduct of post occupancy survey for occupants satisfaction on energy and environmental practices.

Required number of people surveyed shall be:

- 10% of total occupancy and up to 100 maximums.
- Minimum 5 people shall be surveyed if total occupancy is less than 50.

3-2(b)

1 credit can be scored for the list of corrective actions taken following the post evaluation.

DOCUMENTARY EVIDENCES

For 3-2(a)

- Draft of survey form.
- Letter of commitment to conduct the survey.

For 3-2(b)

- List of corrective action will be taken if the feedback of the survey is not satisfying.
- Evidence from past feedback that improvements were implemented or justification of measures to be taken to improve the current situation.

ENRB 3-3 WASTE MANAGEMENT

OBJECTIVES

To promote and encourage recycling and waste minimisation within the occupants of the building.

APPLICABILITY

Generally applicable to all building developments.

REQUIREMENTS

3-3(a)

2 credits for the provision of facilities or recycling bins for collection and storage for different recyclable waste such as paper, glass, plastic, food waste, etc.

For commercial building, waste separation bins to be provided at each floor in a convenient location and suitable strategy should be available to manage the waste collection centrally

3-3(b)

2 credits for promoting and encouraging minimization and recycling among occupants, tenants and visitors through various avenues.

3-3(c)

1 credit for providing proper storage area for recyclable waste.

3-3(d)

2 credit for quantifying and monitoring the recycling programme for continuous improvement.

DOCUMENTARY EVIDENCES

For 3-3(a)

- Layout plan showing the collection and storage for the different recyclable waste.

For 3-3(b)

- Draft of promotion and encouragement plan for minimization and recycling among occupants.

For 3-3(c)

- Layout plan showing the location of the storage area for recyclable waste.

For 3-3(d)

- Waste management plan

ENRB 3-4 SUSTAINABLE PRODUCTS

OBJECTIVES

To promote use of environmentally friendly products that are certified by approved local certification body and are applicable to non-structural and architectural related building components.

APPLICABILITY

Generally applicable to all building developments.

REQUIREMENTS

Up to 8 credits are allocated to encourage the use of environmentally friendly products that are certified by approved local/international certification body. The criterion is only applicable for non-structural building components and construction. Credits scored will be based on the extent of use of environmentally friendly product.

The environmentally friendly product proposed must be approved by a valid international or local certification body and is subject to GreenRE's evaluation.

Extent of use of environmentally friendly product	Weightage for Credits Allocation
Low impact	0.5
Medium Impact	1
High Impact	2

Table 3-2(a) : Weightage for credits allocation

The use of environmental friendly products or recycled materials used for all dwelling units of the development will be considered as high impact (2 credits) on condition that quantities used by percentage are more than 50% (i.e extent of coverage as compared to total quantities used for same intended purpose. If not met, it will be classified as medium impact (1 credit).

Items that are used for all common areas, external works and communal facilities are considered as medium impact (1 credit) if quantities used by percentage are more than 80% (i.e extent of coverage as compared to total quantities used for same intended purpose in common areas If not met, it will be classified as low impact (0.5 credit)

Note:
1. The impact categories listed above generally apply to main building elements – i.e internal / external wall, floor, ceiling, roof, doors, etc. Singular products – i.e termite treatment system, playground equipment, gym flooring etc will be classed as low impact. All applications will be subject to GreenRE's evaluation.
2. Same type of the product not allowed to be double claimed for different area application
3. The credit allocated for low volatile organic compound (VOC) paints and adhesives certified by approved local certification body can be found in ENRB 4-2 and hence shall not be included in the scoring for ENRB 3-2.

DOCUMENTARY EVIDENCES

- Extracts from the tender specification and drawings showing the requirements to incorporate the environmental friendly products that are certified and approved by local/international certification body.
- Certification details from approved local/international certification body such as the material certification standards and rating within validity period.
- Technical product information on the sustainable products.
- Calculation of products and extent of coverage.

REFERENCES

1. http://www.sirim-qas.com.my/index.php/zh/our-services/product-certification/eco-labelling-scheme

2. http://www.sec.org.sg/sgls/

WORKED EXAMPLE 3-4

Determine if the environmentally friendly products selected are certified with approved local/international certification body. Check if the products used are meant for main building elements or functional spaces and can be considered high impact or low impact. Products that are meant for common areas and external works such as toilets, lobbies and landscaping areas are considered as medium impact or low impact.

Note: Certain products can have more environmentally friendly features than others. Other than recycled materials, they may have features like low VOC assembly or manufactured with resource efficient processes, durability etc that will render the products more environmental friendly than others. If the certified products selected are more environmental friendly and are given a better rating by the approved local/international certification body, a higher weightage can be considered in credit scoring.

Example of a proposed development with the following provisions:

a. Use of carpets for all office spaces. Product is not certified.

b. Use of panel boards as internal partitions for more than 50% of the office spaces and the product is rated by an approved certification body.

c. Precast concrete road kerbs. Product is rated by approved local certification body. – (Singular product)

d. Use of roof waterproofing coating. Product is rated by approved local certification body.

e. Use of wooden doors for all areas. Product is rated by approved local certification body.

Products and Extent of coverage		With approved certification	Extent of use category	Credits scored
a.	Carpets for all office spaces	No	N/A	0
b.	Panel boards as internal partition for more than 50% of office spaces	Yes	2	2
c.	Precast road kerbs	Yes	0.5	0.5
d.	Roof waterproofing	Yes	1	1
e.	Wooden doors for all areas	Yes	2	2

Table 3-2(b) : Detail calculation for credits scoring

Therefore, credits scored for 3-2 = 2 + 0.5 + 1 + 2 = 5.5 credits

ENRB 3-5 GREENERY PROVISION

OBJECTIVES

Encourage greater use of greenery and restoration of existing trees to reduce heat island effect.

APPLICABILITY

Applicable to building developments with landscaping areas.

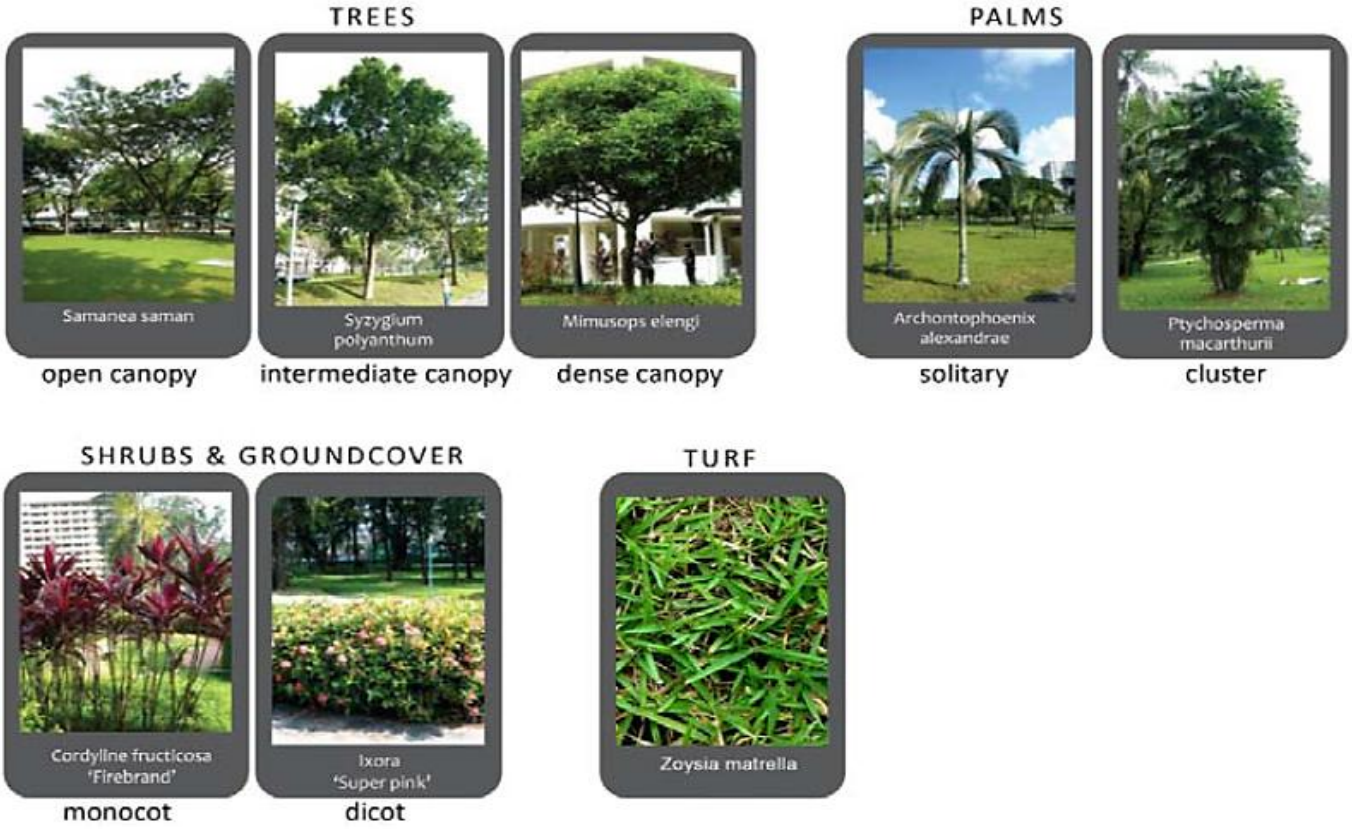
REQUIREMENTS

3-5(a)

Up to 6 credits can be scored for the provision of greenery within the developments including roof top/ sky garden and green roof.

Green Plot Ratio (GnPR) is calculated by considering the 3D volume covered by plants using the following Leaf Area Index (LAI):

Plant Group	Trees	Palms	Shrubs & Groundcover	Turf
LAI	Canopy: Open = 2.5 Intermediate = 3.0 Dense = 4.0	Solitary = 2.5 Cluster = 4.0	Monocot = 3.5 Dicot = 4.5	Turf = 2.0
Area	All = 60 m^2	Solitary = 20m^2 Cluster = 17m^2	Planted area	Planted area



GnPR	Credits Allocation
1.0 to < 2.0	1
2.0 to < 3.0	2
3.0 to < 4.0	3
4.0 to < 5.0	4
5.0 to < 6.0	5
≥ 6.0	6

Green Plot Ratio (GnPR) = Total Leaf Area / Site Area

3-5 (b)

1 credit for restoration of trees on-site, conservation or relocation of existing trees on site.

3-5 (c)

1 credit for the of compost bins to recycle organic waste to meet at least 30% of landscape fertilizer needs.

DOCUMENTARY EVIDENCES

For 3-3 (a)

- Plan layouts showing the site area as well as the greenery that is provided within the development (including a listing of the number of trees, palms, shrubs, turf and the respective sub category and LAI values).
- Calculation showing the extent of the greenery provision in the prescribed tabulated formats.
- The plant species sub categories and its LAI values obtained from the online website: http://florafaunaweb.nparks.gov.sg/

For 3-3 (d)

- Site layouts showing the existing and final locations (where applicable) and number of the trees to be restored or conserved or relocated.
- Documentary evidence showing the relocation or restoration activities.

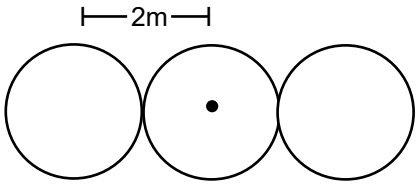
For 3-3 (e)

- Extracts of the tender specification showing the requirements to provide compost bin
- Product specifications.
- Method statement with details steps of composting process (if applicable).
- The calculation of the 30% of fertilizer replacement with the composting

EXCEPTIONS

Trees and Palms Spacing (Centre-to-Centre)

a. If the selected trees and palms are to be planted at ≤ 2m from trunk-to-trunk as illustrated below, the leaf area shall be calculated as the product of LAI value and planted area (in m^2).



Columnar Trees

b. For trees that have tight, columnar crowns, the canopy area of 12m2 is to be adopted for calculation of leaf area. These species include, but not limited to the following:

- Garciniacymosa forma pendula
- Garciniasubelliptica
- Polyalthialongifolia
- Carallia brachiate
- Gnetumgnemon

REFERENCES

The plant species, its sub categories and LAI values may be obtained from the online website: http://florafaunaweb.nparks.gov.sg

WORKED EXAMPLE 3-5(a)

1. Determine the number of trees, palms and the trees for shrubs and turfs and other greenery area.
2. The Leaf Area Index (LAI) of the individual plant species and its canopy area are predetermined design parameters applicable for all developments.
3. The plant species sub categories and its LAI values can be obtained from the online website: http://florafaunaweb.nparks.gov.sg/ (see example below) by searching the common / scientific names of the plants.
4. Compute the green areas as shown in the Table 3-3.3 below

Category	Sub Category	(A)	(B)	(C)	(A)x(B)x(C)
		LAI value	Canopy area	Qty/Planted Area	Leaf Area
Trees (no.)	Open Canopy	2.5	60 m^2	0 no.	0
	Intermediate Canopy	3.0	60 m^2	8 no.	1440
	Dense Canopy	4.0	60 m^2	12 no.	2880
	Intermediate columnar canopy*	3.0	12 m^2	4 no.	144
Palms (no.)	Solitary	2.5	30 m^2	10 no.	750
	Solitary (trunk-to-trunk)	2.5	NA	20 m^2	50
	Cluster	4.0	17 m^2	10 no.	680
Shrubs (m^2)	Monocot	3.5	NA	0 m^2	0
	Dicot	4.5	NA	20 m^2	90
Turf(m^2)	Turf	2.0	NA	90 m^2	180
Vertical Greenery (m^2)	-	2.0	NA	10 m^2	20
				Total Leaf Area:	**6234**

Table 3-5(a) : Calculation of the Green Plot Ratio

Note: Green roof landscaping would be calculated as per illustrated above

Assume site area is 2000 m^2

Green Plot Ratio (GnPR) = total leaf area / site area
= 6234 / 2000 = 3.117 < 4.0

Where GnPR = 3.0 to < 4.0

Therefore, credits scored for 3-5(a) = 3 credits

ENRB 3-6 ENVIRONMENTAL PROTECTION

OBJECTIVES

To adopt a sustainable and environmental-friendly procurement and purchasing policy in the operation and maintenance of the building.

To reduce the potential damage to the ozone layer and the increase in global warming.

APPLICABILITY

Generally applicable to all building developments.

REQUIREMENTS

For 3-6(a)

- 1 credit can be allocated for the adoption of sustainable and environmental-friendly procurement and purchasing policy in the operation and maintenance of the building.

For 3-6(b)

- 1 credit can be scored for the use of refrigerants with ozone depleting potential (ODP) of zero or with global warming potential (GWP) of less than 100.
- 1 credit can be scored for the use of refrigerant leak detection system at critical areas of plant rooms containing chillers and other equipment with refrigerants.

DOCUMENTARY EVIDENCES

For 3-6(a)

- Draft green procurement policy documents to demonstrate environmental preferable services (operation and maintenance).

For 3-6(b)

- Schematic drawing showing the location of the refrigerant leak detection system at critical areas of plant room containing chillers and others equipment with refrigerants.
- Product catalogue of the installed refrigerant leak detection system.

ENRB 3-7 GREEN TRANSPORT

OBJECTIVES

Promote environmental friendly transport options and facilities to reduce pollution from individual car use.

APPLICABILITY

Generally applicable to all building developments.

REQUIREMENTS

3-7(a)

1 credit can be scored for design that provides good access (<800m walking distance) to public transport networks such as MRT/LRT stations or bus stops.

3-7(b)

1 credit can be scored for provision of covered walkway to facilitate connectivity and the use of public transport.

3-7(c)

1 credit can be scored for provision of electric vehicle charging stations and priority parking lots within the development. Provision of minimum of isolator with 7kWp charger

Extent of coverage:

Minimum 1 number priority parking bays for every 100 carpark lots.
EV chargers – 1 for every 200 parking bays. (Cap at 3)

3-7(d)

Up to 1 credit can be scored for the provision of covered/sheltered bicycles parking lots with rack / locking bar.

Extent of Coverage:

Bicycles parking lot: Minimum 10 number and maximum 50 numbers of bicycle parking lot.

Shower Facilities: Minimum 1 number for every 100 regular occupant and additional 1 for every 150 occupants. (Cap at 7)

DOCUMENTARY EVIDENCES

For 3-7(a)

- Site layout plan in the context of the surrounding area showing the location of the development site and walking path to the location of the MRT/LRT stations and bus stops not more than 800m.
- Proposed bus-stop details drawing.

For 3-7(b)

- Site layout plan showing the connection of covered walkway from the development to the MRT/LRT stations or bus stops.
- Extracts of the tender specification showing the requirement to provide covered walkway.

For 3-7(c)

- Extracts of the tender specification showing the requirement to provide hybrid/electric vehicle refuelling/recharge stations and priority parking bays.
- Plan layout showing the location of the electric vehicle charging station in the development.
- Calculation showing numbers of priority parking and charging station to the ratio of overall parking provided.
- Product technical information.

For 3-7(d)

- Extracts of the tender specification showing the requirement to provide covered/sheltered bicycles parking lots for the development and the total quantity of bicycles lots provided.
- Plan layout showing the location of the covered/sheltered bicycle parking lots and rack/locking bar

Other Green Requirements



Part 4 – Indoor Environmental Quality

ENRB 4-1 Indoor Air Quality Performance

ENRB 4-2 Indoor Air Pollutants

ENRB 4-3 Lighting Quality

ENRB 4-4 Thermal Comfort

ENRB 4-5 Internal Noise Level

ENRB 4-1 INDOOR AIR QUALITY PERFORMANCE

OBJECTIVES

To promote a healthy indoor environment for occupant

APPLICABILITY

Generally applicable to all building developments (air-conditioned areas only)

BASELINE STANDARD

Indoor Air Contaminants Parameters:

Physical Parameters	Acceptable Range		
Air Temperature	23 - 26 °C		
Relative Humidity	50 - 70%		
Air Movement	0.15 - 0.5 m/s		
Chemical Contaminants	**Acceptable Limits**		
	ppm	**mg/m³**	**Cfu/m³**
Carbon Monoxide	10	-	-
Formaldehyde	0.1	-	-
Ozone	0.05	-	-
Respirable particulates	-	0.15	-
Total volatile organic compounds (TVOC)	3	-	-
Biological Contaminants	**Acceptable Limits**		
	ppm	**mg/m³**	**Cfu/m³**
Total Bacteria Counts	-	-	500
Total Fungal Counts	-	-	1000
Ventilation Performance Indicator	**Acceptable Limits**		
	ppm	**mg/m³**	**Cfu/m³**
Carbon Dioxide	C1000	-	-

Notes:

- For chemical contaminants, the limits are eight-hour time-weighted average airborne concentrations.
- mg/m^3 is milligrams per cubic meter of air at 25° Celsius and one atmosphere pressure.
- ppm is parts of vapour or gas per million parts of contaminated air by volume.
- cfu/m^3 is colony forming units per cubic meter.
- C is the ceiling limit that shall not be exceeded at any time. Readings above 1000ppm are indication of inadequate ventilation.
- Excess of bacterial counts does not necessarily imply health risk but serve as an indicator for further investigation.

REQUIREMENTS

4-1(a)

Up to 4 credits will be given for conducting a full IAQ audit once every 3 years to comply with the Code of Practice on Indoor Air Quality Department of Occupational Safety and Health, Ministry of Human Resources Malaysia (2005).

4-1(b)

1 credit for the implementation of effective IAQ management plan to ensure building ventilation systems are frequently maintained.

4-1(c)

1 credit for the use of high efficient filter (at least MERV 8) in AHU to reduce indoor contaminations and provide good protection for cooling coil and reducing frequency or eliminating duct cleaning

4-1(d)

1 credit for providing room temperature display (at least 1 unit per floor).

4-1(e)

1 credit for additional carbon dioxide sensor display (at least 1 unit per Floor).

PREREQUISITE

A formal IAQ management plan must be established and implemented to ensure a healthy indoor environment for all occupants. The policy should outline procedures for maintaining acceptable indoor air quality, including ventilation, pollutant source control, and regular maintenance of HVAC systems. The policy should contain at least the following:

- **Ventilation Management**: Ensure adequate ventilation rates in accordance with recognized standards (e.g., ASHRAE 62.1 or local equivalents) to dilute indoor pollutants and maintain fresh air supply.
- **Pollutant Source Control:** Identify and minimize sources of indoor air contaminants, including volatile organic compounds (VOCs), particulate matter, and biological pollutants. This includes the selection of low-emission materials and proper storage of chemicals.
- **HVAC System Maintenance:** Establish a routine inspection and maintenance schedule for HVAC systems to ensure optimal performance, cleanliness, and efficiency. Filters should be replaced regularly, and air ducts cleaned as needed.
- **Monitoring and Response:** Implement procedures for periodic IAQ monitoring and establish a response procedure for occupant complaints or IAQ-related incidents.
- **Occupant Awareness:** Promote awareness and education among building users regarding IAQ practices, including the importance of reporting issues and maintaining cleanliness.



DOCUMENTARY EVIDENCES

For 4-1(a):

- Most recent IAQ audit report highlighting the parameters that contribute to indoor air quality performance.
- Most recent IAQ assessment report with the results of the building air quality.
- Letter of commitment to conduct IAQ assessment IF the assessment not conduct previously.

For 4-1(b):

- Provision of IAQ Management Plan which evaluates overall building ventilation system using checklist and any comparable methods.

For 4-1(c):

- Technical product specification of the filter that is implemented in the AHU.
- Layout plan showing the location of the AHU in the building.

For 4-1(d):

- Layout plan for every floor showing the location of the room temperature displays.
- Technical specification of the temperature displays.

For 4-1(e):

- Layout plan for every floor showing the location of the carbon dioxide sensor display of the building.
- Technical product specification of carbon dioxide sensor.

REFERENCES

Code of Practice on Indoor Air Quality Department of Occupational Safety and Health, Ministry of Human Resources Malaysia.

ENRB 4-2 INDOOR AIR POLLUTANTS

OBJECTIVES

Minimise airborne contaminants, mainly from inside sources to promote a healthy indoor environment.

APPLICABILITY

Generally applicable to building developments.

REQUIREMENTS

4-2(a)

1 credit can be scored for the use of low volatile organic compounds (VOC) paints certified under local / international certification body for at least 90% of the internal wall areas.

4-2(b)

1 credit can be scored for the use adhesives certified under local / international certification body in all composite wood products used for the development.

DOCUMENTARY EVIDENCES

For 4-2(a)

- Extracts of the tender specification showing the requirement to use low VOC paints that are certified by approved local/ international certification body or equivalent.
- Product catalogue.
- Product certificate with validity expiry.

For 4-2(b)

- Extracts of the tender specification showing the requirement to use adhesive with low emission formaldehyde and are certified by approved local/ international certification body.
- Product catalogue.
- Product certificate with validity expiry

ENRB 4-3 LIGHTING QUALITY

OBJECTIVES

To encourage good workplace lighting quality to promote productivity and occupants' comfort

APPLICABILITY

Generally applicable to all building developments.

BASELINE STANDARD

Luminance level stated in MS 1525:2019 – Energy Efficient and use of renewable energy for non-residential buildings – Code of Practice.

REQUIREMENTS

4-3(a) Lighting level to comply with MS 1525:2019

- 1 credit will be provided if the lighting level comply with MS 1525:2019.

4-3(b) Controllability of the lighting system

- Up to 2 credits will be given if at least 90% of the occupants are able to adjust lighting to suit their need and preference.

Controlled by light switches	1 credit
Controlled by task lights	2 credits

4-3(c) High Frequency Ballast

Up to 2 credits will be given if all applicable areas in the entire building are served with fluorescent tubes using high frequency ballast.

20% to < 40%	0.5 credit
40% to < 60%	1 credit
60% to < 80%	1.5 credits
80% and above	2 credits

DOCUMENTARY EVIDENCES

For 4-3(a)

- Current lighting schedule showing lighting levels in various building areas.
- Lighting layout plan.
- Lux simulation showing the lighting level comply with MS1525:2019.
- Technical product information of the lighting luminaries used

For 4-3(b)

- Schematic drawing showing the location of controllable switches / task lights.
- Layout plan showing the coverage of each switch.

For 4-3(c)

- A summary sheet listing all fluorescent luminaries/LED lighting used for the developments and those with high frequency ballasts/LED driver.
- Technical product information of high frequency ballasts/LED driver in all lighting luminaries used.
- Product catalogue of the light fittings used.
- Electrical lighting layout indicating all the fittings with high frequency ballast/LED driver.

REFERENCES

Luminance level stated in MS 1525:2019 - Energy Efficient and use of renewable energy for non-residential buildings – Code of Practice.

ENRB 4-4 THERMAL COMFORT

OBJECTIVES

Recognise buildings that are designed with good thermal comfort.

APPLICABILITY

Generally applicable to all building developments with air-conditioning systems.

BASELINE STANDARD

- Indoor dry-bulb temperature within 23°C to 26°C
- Relative humidity between 50% to 70%

REQUIREMENTS

4-4(a)

1 credit can be scored by ensuring the consistent indoor conditions for thermal comfort:

4-4(b)

1 credit can be scored by giving the flexibility of temperature controllability.

DOCUMENTARY EVIDENCES

For 4-4(a)

- Commissioning report of indoor dry-bulb temperature and relative humidity achieved consistent indoor thermal comfort for at least 48 hours.
- Demonstrate compliance with the committed design specifications.

For 4-4(b)

- Layout plan showing the location of the temperature control equipment.
- Product catalogue of temperature control device installed.

REFERENCES

"Code of Practice on Indoor Air Quality" (2005), Department of Occupational Safety and Health, Ministry of Human Resources Malaysia

ENRB 4-5 INTERNAL NOISE LEVEL

OBJECTIVES

Recognise buildings that are designed to consider the potential noise levels within the dwelling units are maintained at an appropriate level. All building partitions to shall be in accordance with required STC ratings.

APPLICABILITY

Generally applicable to building developments.

BASELINE STANDARD

ASTEM E413 or equivalent

REQUIREMENTS

1 credit can be scored if the building is designed to achieve ambient internal noise level as specified:

- 55dB (6am – 10pm) LAeq
- 45dB (10pm – 6 am) LAeq

This can be achieved by adhering to the following STC values for residential building partitions.

Description	Sound Transmission Class (STC)
Separation between functional spaces within dwelling units and in-between adjacent dwelling units.	40 - 50
Spaces between mechanical and equipment spaces and occupied spaces	50 - 60

For developments that are in close proximity to road with heavy traffic, flyover or highway, it is necessary to have a detailed analysis conducted by the acoustic consultant. Credits can only be scored if the recommendations from the acoustic consultant are implemented.

DOCUMENTARY EVIDENCES

- Extracts of the tender specification showing the requirement to design the occupied space with partitions meeting the required STC ratings as per table below:

Location	STC rating of partitions
Between General Office Space	40 - 50
Hotel Rooms, Classrooms, Lecture Theaters, Meeting Rooms, Conference Rooms and spaces where confidential speech is required	50 - 60
Between Mechanical / Equipment spaces and occupied spaces	50 - 60

- Architectural & structural plan layout, elevation and sectional plans showing types of wall system used, dimensions and size of all building and structural elements with STC ratings.

OR

- A report of detail analysis and recommendations from acoustic consultant (if applicable).

Other Green Requirements



Part 5 – Other Green Features

ENRB 5-1 GREEN FEATURES & INNOVATIONS

OBJECTIVES

Encourage the use of green features that are innovative and have positive environment impact on water efficiency, environment protection and indoor environment quality of the buildings.

APPLICABILITY

Generally applicable to all building development.

REQUIREMENTS

Up to 10 credits are awarded for the use of the following green features depending on their potential environmental benefits or reduced environmental impacts.

Water efficiency

i. Use of self-cleaning façade system
   - 2 credits for more than 75% of the external walls.
   - 1 credit for more than 50% of the external walls.
   - 0.5 credit for at least 25% of the external walls.

ii. Use of grey water recycling system
   - 2 credits for all blocks of the development.
   - 1 credit for at least one block of the development.

iii. Recycling of AHU condensate
   - 1 credit for more than 75% of the AHU condensate
   - 0.5 credit for at least of 50% of the AHU condensate

iv. 0.5 credit for the use of non-chemical water treatment for cooling tower.

Environmental Protection

i. Provision of green roof and roof top garden
   - 1 credit for more than 50% of the roof areas
   - 0.5 credit or at least 25% of the roof areas

ii. Provision of vertical greening
   - 1 credit for more than 50% of the external wall areas
   - 0.5 credit for at least 25% of the external wall areas

iii. 1 credit for the provision of double refuse chutes for separating recyclable from non-recyclable waste.

iv. 0.5 credit for the use of non-chemical treatment system such as termite baiting system, anti-termite mesh.

Indoor Air Quality

i. Use of Titanium Dioxide solutions to remove odour in toilets:
   - 1 credit for more than 50% of all toilets
   - 0.5 credit for at least 25% of all toilets

ii. 1 credit for the use of pneumatic waste collection system.

iii. 0.5 credit for the use of Ultraviolet light-C band (UV) emitters in all air handing units (AHUs) to improve indoor air quality.

iv. Demonstrating the external view in the net lettable area (NLA). The submission must be showing the furniture plan layout.
   - 1 credit for more than 60% of the NLA having the external view
   - 2 credit for more than 75% of the NLA having the external view.

Others

i. Provision of landscape drainage and infiltration trenches:
   - 1 credit for at least 25% of the green areas
   - 0.5 credit for less than 25% of the green areas

ii. Provision of system to recycle surface runoff from the vertical green wall and sky garden:
   - 1 credit for at least 25% of green areas
   - 0.5 credit for less than 25% green areas

iii. 0.5 credit for the use of siphonic rainwater discharge system at roof.

iv. 0.5 credit for the provision of eco-pond.

v. 0.5 credit for the provision of carpark guidance system.

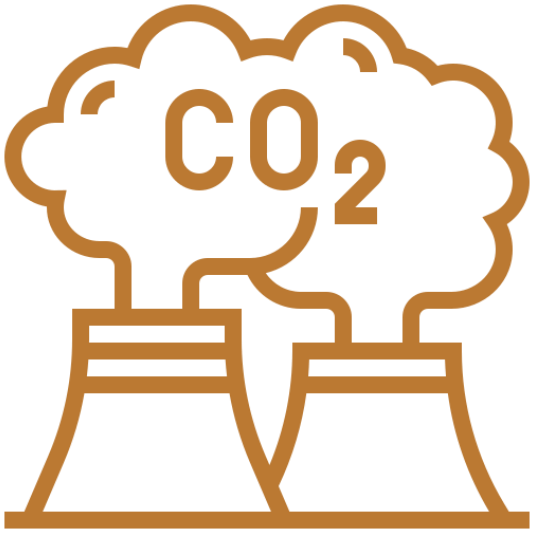
Note: For features that are not listed above, the QP is required to submit the details showing the positive environmental impacts, possible savings and benefits of the proposed features to GreenRE for assessment.

DOCUMENTARY EVIDENCES

- As-built drawings showing the installed green features.
- Product catalogue of the installed green features and its delivery order. A summary sheet listing the breakdown and the extent of implementation as well as the total requirements for the same intended purpose for the specific green features used.
- A summary sheet listing the breakdown and the extent of implementation as well as the total requirements for the same intended purpose for the specific green features used.
- Quantified evidence on the potential environmental benefits that the features can bring to the development.

Other Green Requirements

Part 6 – Carbon Emission of Development



ENRB 6-1 CARBON EMISSION OF DEVELOPMENT

OBJECTIVES

To calculate the carbon emission resulted from the associated energy used during construction and operational phase of a development.

APPLICABILITY

Generally applicable to all building development.

REQUIREMENTS

a. Recognise the carbon emission based on operational carbon footprint computation of the building comprising energy [B6] and water consumption [B7].

b. Calculation of product stage embodied carbon based on following building materials [A1-A3]:

- Concrete
- Steel
- Bricks
- Cement
- Others : Aluminium / Glass / Timber

DOCUMENTARY EVIDENCES

For 6-1 (a)

- Detail calculation for the estimated energy load for each component in the building e.g.: lighting, air-conditioning system, pump, receptacle load.
- Details calculation for estimated water consumption of the building e.g.: water fittings, landscape, water features.
- Technical product information on the energy efficient features and water efficient features used.
- Summary tabulation of estimated total energy savings and total water savings of the development for the year.
- Carbon emission calculation.

For 6-1 (b)

- Embodied carbon footprint calculation

WORKED EXAMPLE 6-1

6-1 (a) Operational Carbon Calculation

Table 6-1: Energy Consumption

Type of usage	Design	Baseline
	(kWh/yr)	(kWh/yr)
Lighting	819,498	1,151,575
Air-Conditioning	860,589	1,406,899
M/V System	25,550	25,550
Total Energy Usage	1,705,637	2,584,024

Table 6-1.2: Water Consumption

(Please refer GreenRE Water Calculator)

Type of fixtures	Design	Baseline
	(m^3/yr)	(m^3/yr)
Flow Fixtures	2,402	6,899
Flush Fixtures	5,366	5,161
Total Water Usage	7,768	12,060

Table 6-1.3: Operational Carbon Footprint

Type of usage	Design	Baseline
	$kgCO_2e/yr$	$kgCO_2e/yr$
Energy	1,226,619	1,860,497
Water	155,344	241,192
Total Annual Carbon Footprint	1,381,963	2,101,689

Energy CO_2 Emission Reduction ($ktCO_2e$/annum, based on electricity energy reduction only @ 1kWh =

- 0.639 kg CO_2- Peninsular
- 0.512 kg CO_2- Sarawak
- 0.512 kg CO_2- Sabah

Water CO_2 Emission Reduction = 0.419 kg CO_2e/m3

Percentage savings = (2,101,689 - 1,381,963) / 2,101,689 = 34.25%

Credits scored for 6-1 (a) = 1 credit

6-1 (b) Product Stage Embodied Carbon Calculation (A1-A3)

No	Material	Description	Embodied Carbon Factor ($kgCO_2e/kg$)	Quantity		Quantity (kg)	Total Embodied Carbon (tCO_2e)
				Value	Unit		
1	Concrete (G30)	Slab	0.13	5,047	m^3	12,112,800	1,574.66
2	Glass	10mm of Glass (Single Glass excluding Frame)	1.44	12,415	kg	12,415	17.88
3	Steel Reinforcement	Rebar	0.73	247,945	kg	247,945	181.00
4	Brick	Red brick – internal wall	0.21	9,574	m^3	16,850,240	3,538.55
5	Cement	Portland cement	0.91	6,217	m^3	9,748,256	8,870.91
6	Aluminium	Window & door frame	0.93	3,972	kg	3,972	3.69
	Total						**14,186.69**

Table 6-1.4: Tabulation of total embodied carbon of material for product stage (A1-A3)
Source: CIDB- Embodied Carbon Inventory Data For Construction Materials

Credits scored for 6-1 (b) = 1 credit

8. Documentation Requirements

All documents submitted for the REHDA GreenRE Assessment should be duly verified and signed by the Qualified Person (QP) and appropriate practitioners where applicable.

The documentation required for ventilation simulation and energy modelling should also be endorsed by the QP and appropriate practitioners as part of the documentary evidences for certification.

GreenRE Criteria	Required Signatories
Part 1 – Energy Efficiency	
ENRB 1-1 Thermal Performance of Building Envelope- OTTV	PA
ENRB 1-2 Air-Conditioning System	PE
ENRB 1-3 Natural Ventilation/ Mechanical Ventilation	PA/PE
ENRB 1-4 Artificial Lighting	PE
ENRB 1-5 Ventilation in Carparks	PA
ENRB 1-6 Ventilation in Common Areas	PA
ENRB 1-7 Lifts and Escalators	PE
ENRB 1-8 Energy Efficient Practices & Features • Heat Recovery Devices • Motion Sensors/ Photo Sensors • Others	 PE PE S
ENRB 1-9 Energy Policy & Management	FM
ENRB 1-10 Renewable Energy	S
Part 2 – Water Efficiency	
ENRB 2-1 Water Usage and Leak Detection	PE/FM
ENRB 2-2 Water Efficient Fittings	PA/FM
ENRB 2-3 Alternative Water Sources	PE
ENRB 2-4 Water Efficiency Improvement Plans	FM
ENRB 2-5 Irrigation System and Landscaping	PE
ENRB 2-6 Water Consumption of Cooling towers	PE
Part 3 – Sustainable Operation & Management	
ENRB 3-1 Building Operation & Maintenance	FM
ENRB 3-2 Post Occupancy Evaluation	FM
ENRB 3-3 Waste Management	FM
ENRB 3-4 Sustainable Products	PA/FM
ENRB 3-5 Greenery Provision	PA
ENRB 3-6 Environmental Protection	PE
ENRB 3-7 Green Transport	PA
Part 4 – Indoor Environmental Quality	
ENRB 4-1 Indoor Air Quality Performance	S
ENRB 4-2 Indoor Air Pollutants	S
ENRB 4-3 Lighting Quality	PE
ENRB 4-4 Thermal Comfort	S
ENRB 4-5 Internal Noise Level	S
Part 5 – Other Green Features	
ENRB 5-1 Green Features & Innovations	S
Part 6 – Carbon Emission of Development	
ENRB 6-1 Carbon Emission of Development	S

Table: Summary Checklist and the Corresponding Signatories for GreenRE Non-Residential Criteria

1. PA refers to Professional Architect, Landscape Architect
2. PE refers to Professional Engineer, Planner and Quantity Surveyor (QS)
3. FM refers to Facility Manager.
4. S refers to Specialist which includes Facilitator, Project Manager, Energy or Sustainable consultant and Commissioning Specialist.